

# **DSE Physics Textbook Knowledge Cards**

## **Compulsory Book 1**

### **101 Temperature and Thermometers**

#### **1.1.1**

Temperature is a measure of the degree of hotness of an object.

#### **1.1.2**

1 Ice point (lower fixed point for Celsius scale):

temperature of pure melting ice at normal atmospheric pressure

2 Steam point (upper fixed point for Celsius scale):

temperature of steam over pure boiling water at normal atmospheric pressure

### **102 Internal Energy, Heat and Heat Transfer**

#### **1.2.1**

The temperature of a body is a measure of the average kinetic energy due to the random motion of the molecules in a body.

#### **1.2.2**

Heat is the energy transferred from one body to another as a result of a temperature difference.

#### **1.2.3**

Conduction is the transfer of heat from the hot part to the cold part of an object, or from a hot object to a cold object in physical contact. Energy (but not matter) is transferred in the process.

#### **1.2.4**

Convection is the process of heat transfer in a fluid (a liquid or a gas) by the actual movement of the fluid itself.

#### **1.2.5**

Radiation is a process of heat transfer which does not require any medium or particles. It takes place in all directions.

### 1.2.6

- dull black surfaces are good absorbers and good radiators of radiation;
- shiny and light-coloured surfaces are poor absorbers and poor radiators of radiation.

### 1.2.7

For a fixed surface and fixed surroundings, increasing the temperature difference increases the magnitude of net radiative heat transfer per unit time.

### 1.2.8

With surface properties and temperatures fixed, a larger surface area increases both emitted and absorbed radiant power.

## 103 Temperature Changes and Heat Capacity

### 1.3.1

$$P = \frac{Q}{t}$$

P: average power (W); Q: energy transferred (J); t: time interval (s).

### 1.3.2

$$Q = C\Delta T$$

$$C = \frac{Q}{\Delta T}$$

C: heat capacity (J K<sup>-1</sup>); Q: heat supplied (J); ΔT: temperature rise (K or °C). No change of state.

### 1.3.3

The heat capacity of a body is the energy needed to raise the temperature of that body by 1 °C.

### 1.3.4

The specific heat capacity of a substance is the energy needed to raise the temperature of 1 kg of the substance by 1 °C.

### 1.3.5

$$c = \frac{C}{m}$$

c: specific heat capacity (J kg<sup>-1</sup> K<sup>-1</sup>); C: heat capacity (J K<sup>-1</sup>); m: mass (kg).

### 1.3.6

$$Q = mc\Delta T$$

$$c = \frac{Q}{m\Delta T}$$

Q: heat supplied (J); m: mass (kg); c: specific heat capacity ( $\text{J kg}^{-1} \text{K}^{-1}$ );  $\Delta T$ : temperature rise (K or  $^{\circ}\text{C}$ ). Constant c; no change of state.

### 1.3.7

In an insulated heat exchange, energy lost by the hotter body equals energy gained by the colder body. Include the container and any change of state if relevant.

### 1.3.8

The total energy of an isolated system remains constant. Energy can be transferred or transformed, but cannot be created or destroyed. A closed system can still exchange energy with its surroundings, so constant total energy additionally requires no net energy exchange.

## 104 Changes of State

### 1.4.1

When a substance melts or vaporizes, it absorbs latent heat.

### 1.4.2

When a substance freezes or condenses, it releases latent heat.

### 1.4.3

$$l = \frac{Q}{m}$$

$$Q = ml$$

l: specific latent heat ( $\text{J kg}^{-1}$ ); Q: energy absorbed or released (J); m: mass changing state (kg). Pure substance at its transition temperature and fixed pressure.

### 1.4.4

Internal energy is the sum of the microscopic kinetic and potential energies of the particles in a body. It excludes the kinetic energy of its overall motion and the potential energy associated with its overall position.

## 105 Gases

### 1.5.1

$$p = \frac{F_{\perp}}{A}$$

p: pressure (Pa);  $F_{\perp}$ : force normal to the surface (N); A: area ( $\text{m}^2$ ).

### 1.5.2

$$p \propto \frac{1}{V}$$

Boyle's law: fixed mass of gas at constant temperature. p: absolute pressure; V: volume.

### 1.5.3

$$pV = \text{constant}$$

$$p_1 V_1 = p_2 V_2$$

Fixed mass of gas at constant temperature; p: absolute pressure; V: volume. Use consistent units in both states.

### 1.5.4

$$\frac{T}{K} = \frac{\theta}{^{\circ}\text{C}} + 273.15 \approx \frac{\theta}{^{\circ}\text{C}} + 273$$

T: thermodynamic temperature;  $\theta$ : Celsius temperature. The textbook uses the rounded offset 273; 273.15 is exact.

### 1.5.5

$$p \propto T$$

Fixed mass and fixed volume; p: absolute pressure; T: temperature in kelvin.

### 1.5.6

$$\frac{p}{T} = \text{constant}$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

Fixed mass and fixed volume; p: absolute pressure; T: temperature in kelvin.

### 1.5.7

$$V \propto T$$

Fixed mass and constant pressure; V: volume; T: temperature in kelvin.

### 1.5.8

$$\frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Fixed mass and constant pressure; V: volume; T: temperature in kelvin.

### 1.5.9

$$pV \propto T$$

Combined gas law for a fixed mass: p is absolute pressure, V is volume, and T is in kelvin. Neither pressure nor volume must be held constant.

### 1.5.10

$$\frac{pV}{T} = \text{constant}$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Fixed amount of ideal gas; p: absolute pressure; V: volume; T: kelvin temperature.

### 1.5.11

$$pV = nRT$$

p: pressure (Pa); V: volume (m<sup>3</sup>); n: amount (mol); T: temperature (K); R = 8.31 J mol<sup>-1</sup> K<sup>-1</sup>. Ideal gas.

### 1.5.12

Gas pressure is produced when gas molecules collide with a surface.

### 1.5.13

- 1 All the molecules are identical and have the same mass.
- 2 All the molecules are in continual random motion.
- 3 The number of molecules in the container is large.
- 4 The size of each molecule is negligible compared with the separation between them.
- 5 The duration of a collision is negligible compared with the time between collisions.
- 6 The collisions between molecules and the container and between the molecules are perfectly elastic.
- 7 Intermolecular forces are negligible.

### 1.5.14

$$pV = \frac{1}{3}Nm\overline{c^2}$$

p: pressure (Pa); V: volume (m<sup>3</sup>); N: number of molecules; m: mass of ONE molecule (kg). The overbar over  $c^2$  means the mean square speed (m<sup>2</sup> s<sup>-2</sup>), not the square of the mean speed. Ideal gas with isotropic random motion.

### 1.5.15

$$\langle K \rangle = \frac{1}{2}m\overline{c^2} = \frac{3RT}{2N_A}$$

$\langle K \rangle$ : mean translational kinetic energy per molecule (J); m: one molecule's mass (kg); mean square speed: m<sup>2</sup> s<sup>-2</sup>; T: K; R = 8.31 J mol<sup>-1</sup> K<sup>-1</sup>; N<sub>A</sub> = 6.02 × 10<sup>23</sup> mol<sup>-1</sup>.

### 1.5.16

$$T \propto \langle K \rangle$$

For an ideal gas, T is absolute temperature (K), and  $\langle K \rangle$  is mean translational kinetic energy per molecule (J).

# Compulsory Book 2

## 201 Motion

### 2.1.1

the change in position of the object is called displacement.

### 2.1.2

A scalar is described by magnitude only.

A vector is described by both magnitude and direction.

### 2.1.3

Speed = distance travelled per unit time

### 2.1.4

$$v_{\text{speed,av}} = \frac{d_{\text{total}}}{\Delta t}$$

Average speed ( $\text{m s}^{-1}$ ) = total distance travelled (m) / total elapsed time (s). Scalar; include rest intervals.

### 2.1.5

Velocity = displacement per unit time

### 2.1.6

$$\vec{v}_{\text{av}} = \frac{\Delta \vec{s}}{\Delta t}$$

Average velocity ( $\text{m s}^{-1}$ );  $\Delta s$ : resultant displacement (m);  $\Delta t$ : total time (s). State the direction.

### 2.1.7

Acceleration = change in velocity per unit time

### 2.1.8

For uniform motion:

- speed is constant;
- the direction of motion is unchanged, so velocity is constant;
- acceleration is zero.

### 2.1.9

$$\text{percentage uncertainty} = \frac{\delta x}{x} \times 100\%$$

Textbook wording: percentage error = error / measured value  $\times 100\%$ . Here  $\delta x$  is the absolute measurement uncertainty and  $x$  the measured value in the same units. Deviation from a reference value uses that reference in the denominator.

## 202 Motion Along a Straight Line

### 2.2.1

$$a_{\text{av}} = \frac{v - u}{\Delta t}$$

$u$ ,  $v$ : initial and final velocity ( $\text{m s}^{-1}$ );  $\Delta t$ : time (s);  $a_{\text{av}}$ : average acceleration ( $\text{m s}^{-2}$ ). Signed values along one chosen axis; equals instantaneous acceleration when constant.

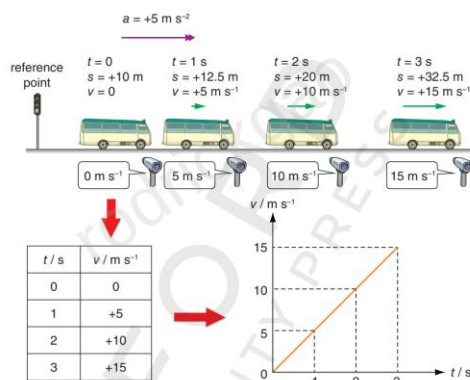


Fig 2.21 Plotting the  $v$ - $t$  graph of a car in uniformly accelerated motion.

Equal increases in velocity in equal times produce a straight  $v$ - $t$  graph. Its slope is acceleration; its signed area gives displacement.

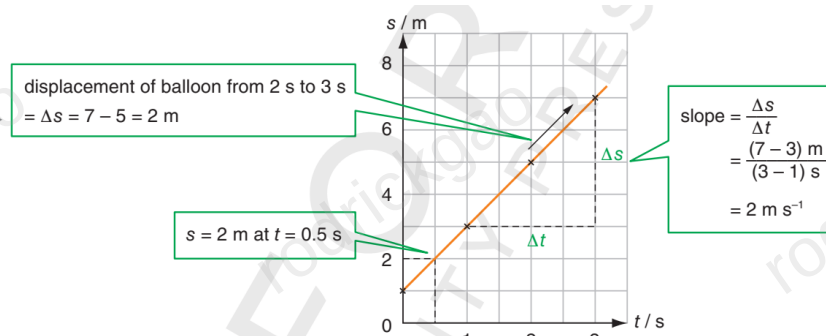
### 2.2.2

- The sign of acceleration of an object does not tell the direction of motion, nor whether the object is speeding up or slowing down.
- The object is speeding up if its velocity  $v$  and acceleration  $a$  have the same sign; it is slowing down if  $v$  and  $a$  have opposite signs.

### 2.2.3

$$v = \frac{ds}{dt}$$

Velocity is the tangent slope of a displacement–time graph. Over an interval  $\Delta s/\Delta t$  gives average velocity.  $s$ : m;  $t$ : s;  $v$ :  $\text{m s}^{-1}$ .



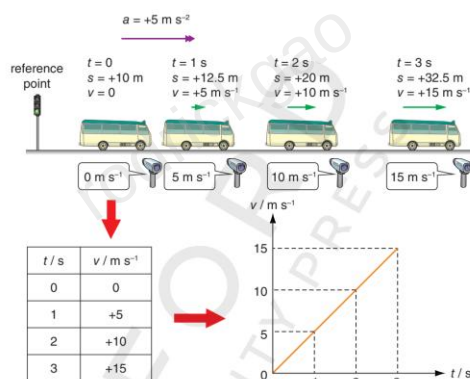
**Fig 2.2b** How position ( $s$ ), displacement ( $\Delta s$ ) and average velocity ( $\Delta s/\Delta t$ ) are found from an  $s$ – $t$  graph.

The slope is  $\Delta s/\Delta t$ : the marked rise of 4 m in 2 s gives  $2 \text{ m s}^{-1}$ . Position is read from the vertical axis; displacement is a change in position.

### 2.2.4

$$a = \frac{dv}{dt}$$

Acceleration is the tangent slope of a velocity–time graph. Over an interval  $\Delta v/\Delta t$  gives average acceleration.  $v$ :  $\text{m s}^{-1}$ ;  $t$ : s;  $a$ :  $\text{m s}^{-2}$ .



**Fig 2.2i** Plotting the  $v$ – $t$  graph of a car in uniformly accelerated motion.

Equal increases in velocity in equal times produce a straight  $v$ – $t$  graph. Its slope is acceleration; its signed area gives displacement.

### 2.2.5

$$\Delta s = \int_{t_1}^{t_2} v \, dt$$

Displacement equals the signed area under a velocity–time graph. Area below the time axis is negative; distance uses the sum of absolute areas. Integral notation represents this area rule.

### 2.2.6

$$s \xrightarrow{\text{slope}} v \xrightarrow{\text{slope}} a$$

$$\Delta s = \int_{t_1}^{t_2} v \, dt$$

$$\Delta v = \int_{t_1}^{t_2} a \, dt$$

For graphs against time: slopes give velocity then acceleration; signed areas give changes in velocity or displacement. Initial values are needed to recover  $v$  or  $s$ .

$$s-t \text{ graph} \xrightleftharpoons[\text{area}]{\text{slope}} v-t \text{ graph} \xrightarrow{\text{slope}} a-t \text{ graph}$$

Read slopes from displacement to velocity to acceleration. Read signed areas in reverse to find changes, adding the appropriate initial value.

### 2.2.7

$$v = u + at$$

$$s = \frac{u + v}{2} t$$

$$s = ut + \frac{1}{2} at^2$$

$$v^2 = u^2 + 2as$$

Constant acceleration along one axis.  $u, v$ : velocities ( $\text{m s}^{-1}$ );  $a$ : acceleration ( $\text{m s}^{-2}$ );  $s$ : displacement ( $\text{m}$ );  $t$ : time ( $\text{s}$ ). Choose a positive direction consistently.

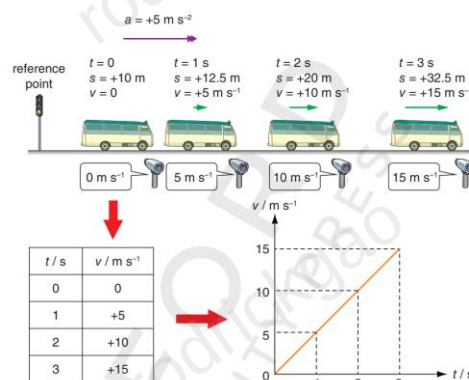


Fig 2.2i Plotting the  $v$ - $t$  graph of a car in uniformly accelerated motion.

Equal increases in velocity in equal times produce a straight  $v$ - $t$  graph. Its slope is acceleration; its signed area gives displacement.

### 2.2.8

$$g \approx 9.81 \text{ m s}^{-2}$$

Near Earth's surface; negligible air resistance. Acceleration is vertically downward, including during upward motion.

### 2.2.9

$$g \approx 9.81 \text{ms}^{-2}$$

Near Earth's surface; negligible air resistance. Acceleration is vertically downward, including during upward motion.

## 203 Force and Motion (I)

### 2.3.1

Inertia is the tendency of an object to maintain its state of rest or of constant velocity.

### 2.3.2

Mass is a measure of an object's inertia.

### 2.3.3

Every object remains in a state of rest or uniform motion (i.e. constant velocity) unless it is acted on by a net force.

### 2.3.4

$$a \propto F_{\text{net}} (m \text{ fixed})$$

$$a \propto \frac{1}{m} (F_{\text{net}} \text{ fixed})$$

a: acceleration magnitude ( $\text{m s}^{-2}$ );  $F_{\text{net}}$ : resultant force magnitude (N); m: mass (kg). Acceleration points along the resultant force.

### 2.3.5

$$\sum \vec{F} = m\vec{a}$$

Resultant force (N), not an individual force; m: constant mass (kg); a: acceleration ( $\text{m s}^{-2}$ ). In an inertial reference frame.

### 2.3.6

$$\vec{W} = m\vec{g}$$

W: weight (N); m: mass (kg); g: gravitational field strength ( $\text{N kg}^{-1}$ ). Weight points along the gravitational field.

### 2.3.7

Mass and weight are different:

- Mass measures inertia. It is a scalar measured in kilograms (kg) and does not change with location.
- Weight is the gravitational force acting on an object,  $W = mg$ . It is a vector measured in newtons (N) and changes when  $g$  changes.

### 2.3.8

An object released from rest in a fluid accelerates downward while its speed-dependent resistance grows. It reaches terminal speed when the resultant force is zero. If buoyancy is negligible, drag then equals weight; otherwise drag plus buoyancy equals weight.

### 2.3.9

For every action, there is an equal and opposite reaction. The action and reaction forces act on different interacting objects simultaneously.

## 204 Force and Motion (II)

### 2.4.1

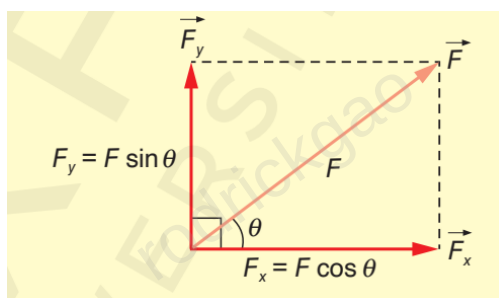
Bodies constrained to move together with the same velocity and acceleration may be analysed as one system. Internal action–reaction forces cancel when finding the resultant force on the whole system; retain them when analysing an individual body.

### 2.4.2

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

$F$ : force (N);  $\theta$ : angle measured from the positive x-axis to  $F$ . Components are signed.



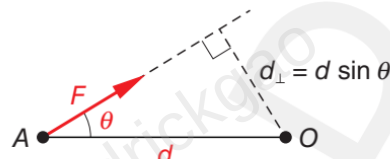
$\theta$  is measured from the horizontal component. The adjacent component is  $F \cos \theta$  and the opposite component is  $F \sin \theta$ .

## 205 Moment

### 2.5.1

$$\tau = Fd_{\perp}$$

$\tau$ : moment (N m);  $F$ : force (N);  $d_{\perp}$ : shortest distance from pivot to force's line of action (m). State clockwise or anticlockwise.



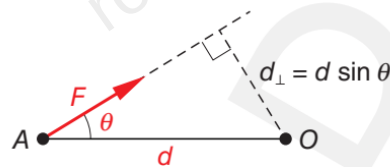
**Fig 5.1c**  $\tau = Fd \sin \theta$ .

The lever arm is the perpendicular distance from O to the force line, not simply OA. Here  $d_{\perp} = d \sin \theta$ .

### 2.5.2

$$\tau = Fd \sin \theta = Fd_{\perp} = F_{\perp} d$$

$\tau$ : moment magnitude (N m);  $d$ : pivot-to-force-application distance (m);  $\theta$ : angle between  $d$  and  $F$ ;  $d_{\perp} = d \sin \theta$ ;  $F_{\perp} = F \sin \theta$ .



**Fig 5.1c**  $\tau = Fd \sin \theta$ .

The lever arm is the perpendicular distance from O to the force line, not simply OA. Here  $d_{\perp} = d \sin \theta$ .

### 2.5.3

$$\tau = Fd \sin \theta = Fd_{\perp} = F_{\perp} d$$

$\tau$ : moment magnitude (N m);  $d$ : pivot-to-force-application distance (m);  $\theta$ : angle between  $d$  and  $F$ ;  $d_{\perp} = d \sin \theta$ ;  $F_{\perp} = F \sin \theta$ .



**Fig 5.1c**  $\tau = Fd \sin \theta$ .

The lever arm is the perpendicular distance from O to the force line, not simply OA. Here  $d_{\perp} = d \sin \theta$ .

### 2.5.4

$$\begin{aligned}\sum \vec{F} &= \vec{0} \\ \sum \tau &= 0\end{aligned}$$

Equilibrium requires zero resultant force and zero resultant moment about any point. Resolve forces and use one consistent sign convention for moments.

### 2.5.5

Every rigid body has a fixed point at which its weight seems to act. The point is called the centre of gravity.

## 206 Work, Energy and Power

### 2.6.1

Work is the energy transferred due to the action of a force over a displacement.

### 2.6.2

$$W = F_{\parallel} s$$

W: work (J);  $F_{\parallel}$ : constant force component along displacement (N); s: displacement magnitude (m). A component opposite the displacement gives negative work.

### 2.6.3

$$W = F s \cos \theta$$

W: work (J); F: constant force (N); s: displacement (m);  $\theta$ : angle between force and displacement.

### 2.6.4

$$1\text{J} = 1\text{Nm}$$

One joule is the work of a one-newton force through one metre along the force. Torque also has unit Nm, but is not energy.

### 2.6.5

$$K = \frac{1}{2} m v^2$$

K (KE): translational kinetic energy (J); m: mass (kg); v: instantaneous speed ( $\text{m s}^{-1}$ ). Non-relativistic motion.

### 2.6.6

$$U = mgh$$

U (PE): gravitational potential energy (J); m: kg; g: N kg<sup>-1</sup>; h: height above the chosen zero level (m).  
Approximately uniform gravitational field.

### 2.6.7

$$\Delta U = mg\Delta h$$

$\Delta U$ : change in gravitational potential energy (J); m: kg; g: N kg<sup>-1</sup>;  $\Delta h$ : final minus initial height (m).  
Constant g.

### 2.6.8

The total energy of an isolated system remains constant. Energy can be transferred or transformed, but cannot be created or destroyed. A closed system can still exchange energy with its surroundings, so constant total energy additionally requires no net energy exchange.

### 2.6.9

$$K_1 + U_1 = K_2 + U_2$$

Mechanical energy conservation when only conservative forces do work. K: kinetic energy; U: potential energy (J).

### 2.6.10

$$K_1 + U_1 = K_2 + U_2 + W_{\text{res}}$$

$W_{\text{res}} \geq 0$ : energy transferred by work against resistance (J). No other energy input; K: kinetic energy; U: potential energy.

### 2.6.11

$$K_1 + U_1 + W_{\text{ext}} = K_2 + U_2$$

$W_{\text{ext}}$ : signed work by external non-conservative forces (J). Do not count gravitational work again when gravitational potential energy is included.

### 2.6.12

$$P_{\text{av}} = \frac{W}{\Delta t}$$

$P_{\text{av}}$ : average power (W); W: work (J);  $\Delta t$ : elapsed time (s).

### 2.6.13

$$P = \vec{F} \cdot \vec{v} = Fv\cos\theta$$
$$P = Fv(\theta = 0)$$

Instantaneous power  $P$  (W);  $F$ : force (N);  $v$ : velocity ( $\text{m s}^{-1}$ );  $\theta$ : angle between them. The textbook's  $P = Fv$  assumes parallel force and velocity.

### 2.6.14

$$K = \frac{1}{2}mv^2$$

$K$  (KE): translational kinetic energy (J);  $m$ : mass (kg);  $v$ : instantaneous speed ( $\text{m s}^{-1}$ ). Non-relativistic motion.

### 2.6.15

$$U = mgh$$

$U$  (PE): gravitational potential energy (J);  $m$ : kg;  $g$ :  $\text{N kg}^{-1}$ ;  $h$ : height above the chosen zero level (m). Approximately uniform gravitational field.

### 2.6.16

$$P = \vec{F} \cdot \vec{v} = Fv\cos\theta$$
$$P = Fv(\theta = 0)$$

Instantaneous power  $P$  (W);  $F$ : force (N);  $v$ : velocity ( $\text{m s}^{-1}$ );  $\theta$ : angle between them. The textbook's  $P = Fv$  assumes parallel force and velocity.

## 207 Momentum

### 2.7.1

$$\vec{p} = m\vec{v}$$

$p$ : momentum ( $\text{kg m s}^{-1}$ );  $m$ : mass (kg);  $v$ : velocity ( $\text{m s}^{-1}$ ). Momentum is a vector; non-relativistic motion.

### 2.7.2

The total momentum of a system is conserved, provided that there is no external net force acting on the system.

### 2.7.3

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

No resultant external impulse during the interaction.  $m_A$ ,  $m_B$ : masses;  $u$ : initial velocity;  $v$ : final velocity. Signed components along one axis; applies to elastic and inelastic collisions.

### 2.7.4

If the total KE of the colliding objects is conserved, the collision is an elastic collision. Otherwise, the collision is an inelastic collision.

### 2.7.5

$$F_{\text{net,av}} = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{\Delta t}$$

Average resultant force (N);  $\Delta p$ : momentum change ( $\text{kg m s}^{-1}$ );  $\Delta t$ : s. Last expression requires constant mass. Instantaneous force is the instantaneous rate of change of momentum.

### 2.7.6

The net force acting on an object is equal to the rate of change in momentum of the object.

### 2.7.7

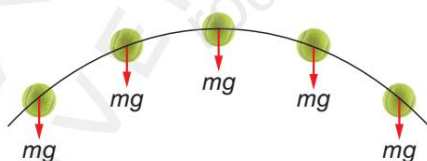
$$\Delta p = \int_{t_1}^{t_2} F_{\text{net}} dt$$

Impulse equals the signed area under the resultant force–time graph ( $\text{N s} = \text{kg m s}^{-1}$ ). Integral notation represents this area rule.

## 208 Projectile Motion

### 2.8.1

- When air resistance is negligible, a projectile moves at a uniform speed horizontally and at a uniform acceleration vertically due to gravity.
- The horizontal motion and vertical motion are independent of each other.



**Fig 8.1c** A projectile is acted on only by the gravitational force  $mg$ .

Gravity acts downward at every point, including the highest point. With negligible air resistance, horizontal velocity stays constant.

## 209 Uniform Circular Motion

### 2.9.1

$$\theta = \frac{s}{r}$$

$\theta$ : angular displacement (rad);  $s$ : arc length (m);  $r$ : radius (m).

### 2.9.2

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Uniform circular motion:  $\omega$  is angular speed ( $\text{rad s}^{-1}$ );  $\Delta\theta$ : swept angle (rad);  $\Delta t$ : time (s). Otherwise the ratio gives average angular velocity.

### 2.9.3

$$v = r\omega$$

$v$ : tangential speed ( $\text{m s}^{-1}$ );  $r$ : radius (m);  $\omega$ : angular speed ( $\text{rad s}^{-1}$ ).

### 2.9.4

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$

Uniform circular motion:  $T$ : period (s);  $\omega$ :  $\text{rad s}^{-1}$ ;  $r$ : m;  $v$ :  $\text{m s}^{-1}$ .

### 2.9.5

$$a_c = \frac{v^2}{r} = r\omega^2$$

Centripetal acceleration  $a_c$  ( $\text{m s}^{-2}$ ) points towards the centre.  $v$ : speed ( $\text{m s}^{-1}$ );  $r$ : radius (m);  $\omega$ :  $\text{rad s}^{-1}$ .

### 2.9.6

$$F_c = \frac{mv^2}{r} = mr\omega^2$$

$F_c$ : inward resultant force (N), not an additional force;  $m$ : kg;  $v$ :  $\text{m s}^{-1}$ ;  $r$ : m;  $\omega$ :  $\text{rad s}^{-1}$ .

## 210 Gravitation

### 2.10.1

$$F = \frac{GMm}{r^2}$$

F: attractive force magnitude (N); M, m: masses (kg); r: centre separation (m);  $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ . Point masses or non-overlapping spherically symmetric bodies.

### 2.10.2

$$g = \frac{GM}{r^2}$$

g: gravitational field strength ( $\text{N kg}^{-1}$ ), numerically free-fall acceleration ( $\text{m s}^{-2}$ ); M: source mass (kg); r: distance from its centre (m). Outside a spherical mass; field points inward.

### 2.10.3

$$\vec{g} = \frac{\vec{F}}{m_0}$$

g: gravitational field strength ( $\text{N kg}^{-1}$ ); F: force on test mass (N);  $m_0$ : test mass (kg).

### 2.10.4

$$g = \frac{GM}{r^2}$$

g: gravitational field strength ( $\text{N kg}^{-1}$ ), numerically free-fall acceleration ( $\text{m s}^{-2}$ ); M: source mass (kg); r: distance from its centre (m). Outside a spherical mass; field points inward.

### 2.10.5

$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$$

Circular orbit: gravity supplies the centripetal force. M: central mass; m: orbiting mass; r: centre-to-centre distance; v: orbital speed;  $\omega$ : angular speed. M is treated as stationary.

### 2.10.6

$$v = \sqrt{\frac{GM}{r}}$$

$$\omega = \sqrt{\frac{GM}{r^3}}$$

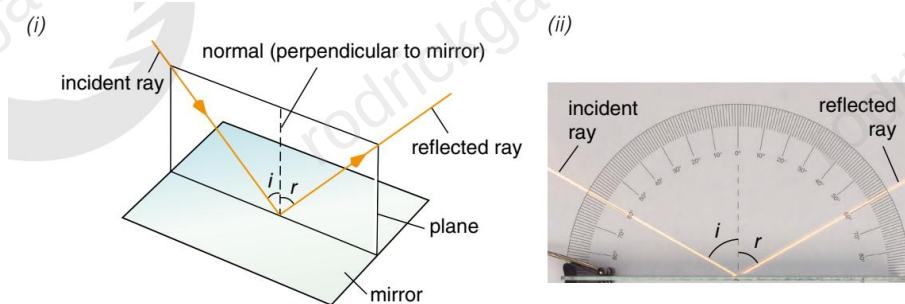
Circular orbit around central mass  $M$  (kg); orbital radius  $r$  (m) is measured from its centre, not the surface.  $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ ;  $v$ :  $\text{m s}^{-1}$ ;  $\omega$ :  $\text{rad s}^{-1}$ .

# Compulsory Book 3

## 3A01 Reflection of Light

### 3.1.1

- The incident ray, the reflected ray and the normal all lie in the same plane.
- The angle of reflection  $r$  is always equal to the angle of incidence  $i$ .



**Fig 1.2c** The laws of reflection.

The incident ray, reflected ray and normal share one plane. Both  $i$  and  $r$  are measured from the normal, giving  $i = r$ .

### 3.1.2

The properties of the image formed by a plane mirror:

- virtual
- behind the mirror
- image distance is equal to object distance
- the same size as the object
- laterally inverted



**Fig 1.2j** The image formed by a mirror is laterally inverted.

The original photograph shows lateral inversion. The virtual image appears behind the mirror, with the same size and perpendicular distance as the object.

## 3A02 Refraction of Light

### 3.2.1

For a light ray travelling from one medium to another at an angle,

- it bends towards the normal if the speed of light decreases;
- it bends away from the normal if the speed of light increases.

Light travels fastest in a vacuum (or air) and slower in other media.

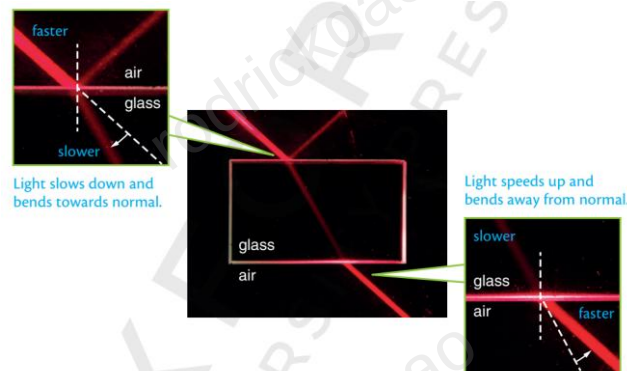


Fig 2.1e Light ray passing through a glass block.

At the air-to-glass boundary light slows and bends towards the normal; on leaving glass it speeds up and bends away. The frequency stays unchanged.

### 3.2.2

$$\frac{\sin i}{\sin r} = \text{constant}$$

Incident ray, refracted ray and normal lie in one plane.  $i$ ,  $r$  are measured from the normal; fixed pair of media and fixed wavelength.

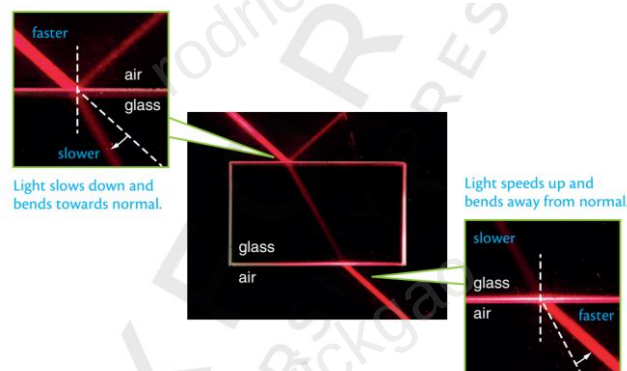


Fig 2.1e Light ray passing through a glass block.

At the air-to-glass boundary light slows and bends towards the normal; on leaving glass it speeds up and bends away. The frequency stays unchanged.

### 3.2.3

$$n_x = \frac{\sin \theta_a}{\sin \theta_x}$$

$n_x$ : refractive index of medium X relative to vacuum (air approximated as  $n = 1$ ). Angles in air and X are measured from the normal.

### 3.2.4

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Snell's law:  $n_1, n_2$  are refractive indices;  $\theta_1, \theta_2$  are angles from the normal, not the surface.

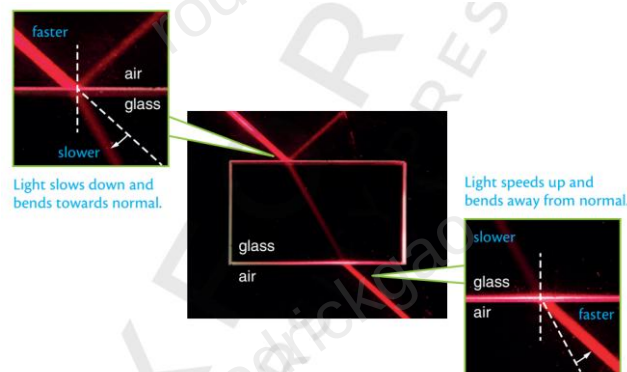


Fig 2.1e Light ray passing through a glass block.

At the air-to-glass boundary light slows and bends towards the normal; on leaving glass it speeds up and bends away. The frequency stays unchanged.

### 3.2.5

$$n = \frac{c}{v}$$

$n$ : refractive index;  $c$ : light speed in vacuum, approximately  $3.00 \times 10^8 \text{ m s}^{-1}$ ;  $v$ : light speed in the medium at that frequency.

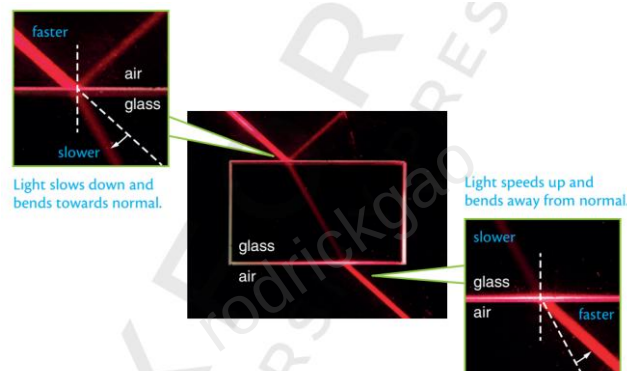


Fig 2.1e Light ray passing through a glass block.

At the air-to-glass boundary light slows and bends towards the normal; on leaving glass it speeds up and bends away. The frequency stays unchanged.

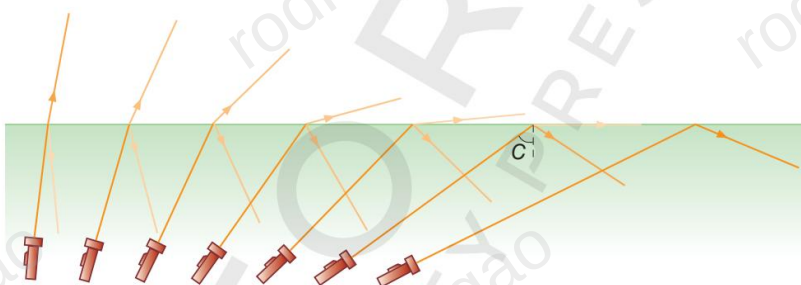
### 3.2.6

Different coloured lights travel at different speeds in a prism and hence have different refractive indices.

### 3.2.7

$$\sin C = \frac{1}{n}$$

C: critical angle measured from the normal; n: refractive index of the denser medium. This form assumes the other medium is air/vacuum; generally  $\sin C = n_2/n_1$  with  $n_1 > n_2$ .



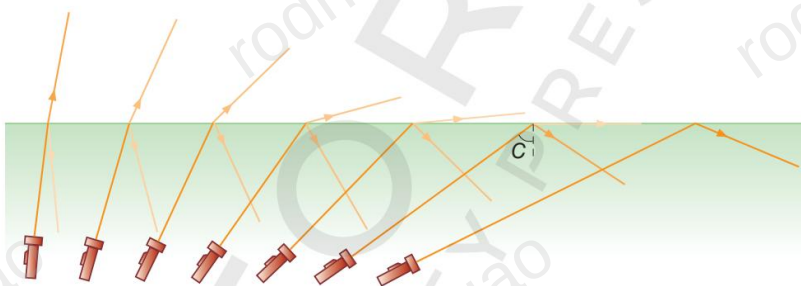
**Fig 2.2b** Total internal reflection occurs only when the angle of incidence is larger than C.

At the critical angle the refracted ray runs along the boundary. Total internal reflection occurs only for incidence greater than C from the higher-index side.

### 3.2.8

Total internal reflection occurs when

- 1 light travels from an optically denser medium to an optically less dense medium, and
- 2 the angle of incidence is larger than the critical angle C.



**Fig 2.2b** Total internal reflection occurs only when the angle of incidence is larger than C.

At the critical angle the refracted ray runs along the boundary. Total internal reflection occurs only for incidence greater than C from the higher-index side.

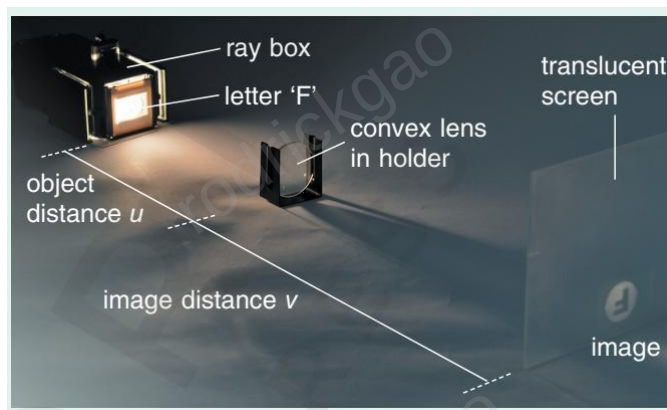
## 3A03 Lenses

### 3.3.1

A convex lens can form virtual images and real images.

No light rays pass through a virtual image. The image cannot be projected on a screen.

Light rays actually pass through a real image. The image can be projected on a screen.

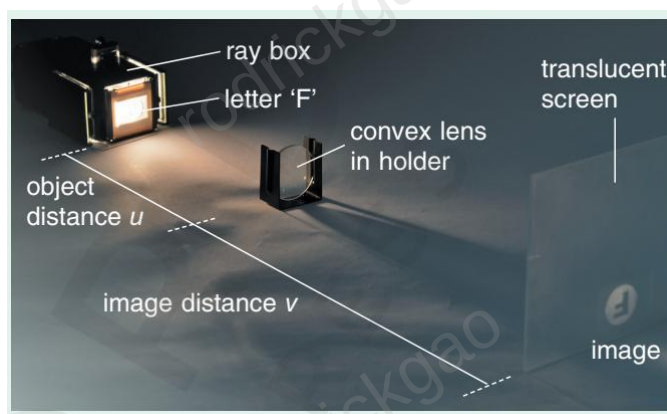


The converging lens forms a real inverted image on the translucent screen. The marked  $u$  and  $v$  are measured from the lens.

### 3.3.2

For a real object a distance  $u$  from a converging lens of focal length  $f$ :

- $u > f$ : a real, inverted image forms on the opposite side.
- $u < f$ : a virtual, upright, magnified image forms on the object side.
- $u = f$ : emergent rays are parallel; no image forms at a finite distance.

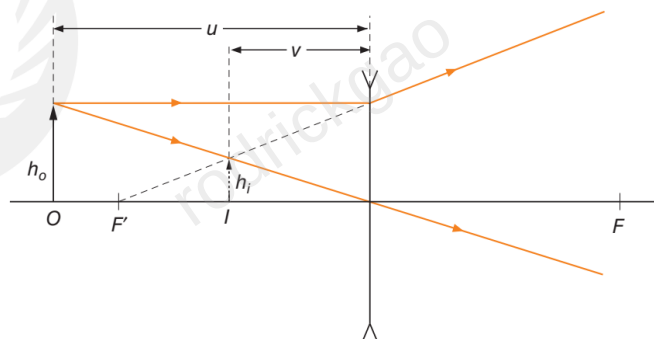


The converging lens forms a real inverted image on the translucent screen. The marked  $u$  and  $v$  are measured from the lens.

### 3.3.3

$$m = \frac{h_i}{h_o} = \frac{|v|}{u}$$

Linear magnification as a size ratio is non-negative.  $h_i$ ,  $h_o$ : image and object heights (same units);  $u > 0$ : real-object distance;  $v$ : signed image distance. The textbook's  $v/u$  uses distance magnitudes here; orientation is stated separately.

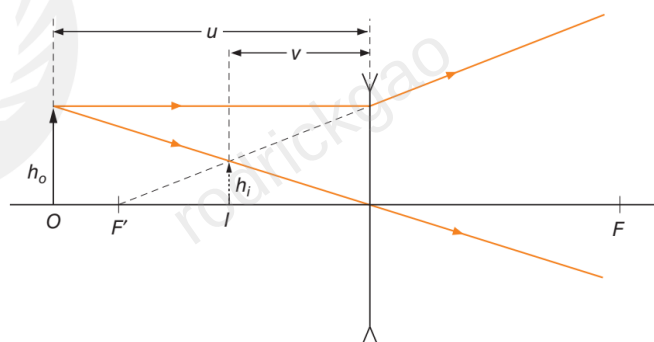


**Fig 3.3e** Linear magnification of the image formed by a concave lens.

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use  $v < 0$ .

### 3.3.4

A diverging lens forms a virtual image for a real object. This statement assumes a real object, rather than a converging incident beam representing a virtual object.

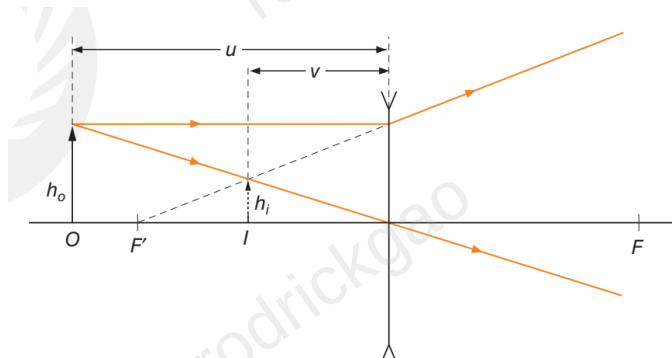


**Fig 3.3e** Linear magnification of the image formed by a concave lens.

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use  $v < 0$ .

### 3.3.5

For a real object, a diverging lens forms a virtual, upright and diminished image on the same side of the lens as the object.



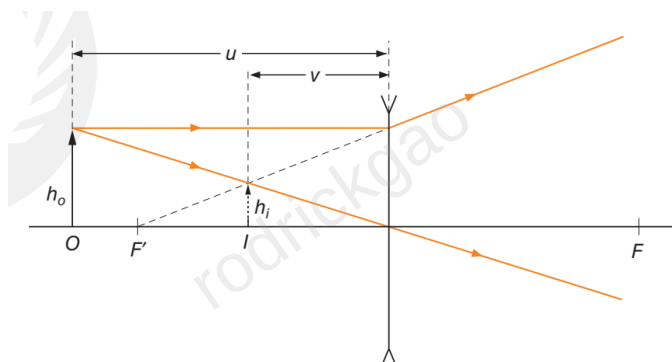
**Fig 3.3e** Linear magnification of the image formed by a concave lens.

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use  $v < 0$ .

### 3.3.6

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Thin lens, paraxial rays. Real object:  $u > 0$ ; real image:  $v > 0$ ; virtual image:  $v < 0$ . Converging lens:  $f > 0$ ; diverging lens:  $f < 0$ . All distances in the same units.



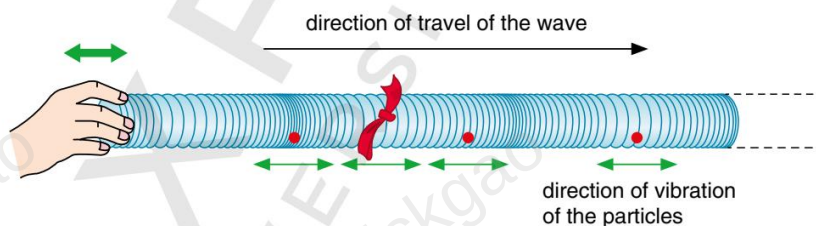
**Fig 3.3e** Linear magnification of the image formed by a concave lens.

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use  $v < 0$ .

## 3B04 Nature of Waves

### 3.4.1

Waves transmit energy without transferring matter.



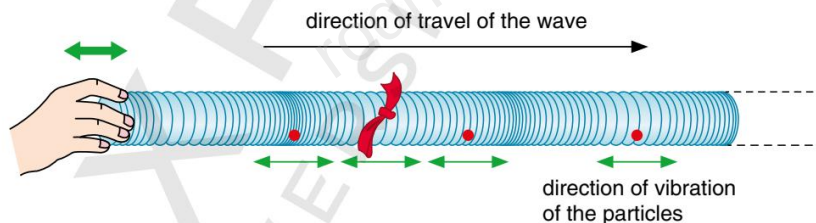
**Fig 4.1i** On a longitudinal wave, particles vibrate along the direction the wave travels.

The particle vibration arrows are parallel to wave travel. Energy propagates while particles oscillate about equilibrium positions.

### 3.4.2

On a transverse wave, the particle vibrations are perpendicular to the direction the wave travels.

On a longitudinal wave, the particle vibrations are parallel to the direction the wave travels.

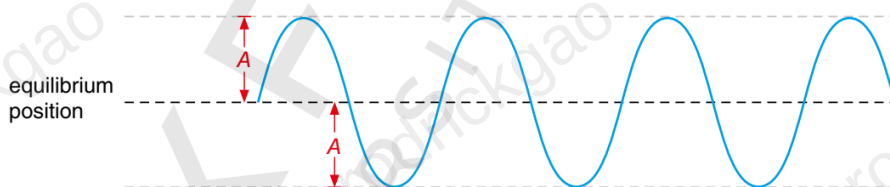


**Fig 4.1i** On a longitudinal wave, particles vibrate along the direction the wave travels.

The particle vibration arrows are parallel to wave travel. Energy propagates while particles oscillate about equilibrium positions.

### 3.4.3

The amplitude of a transverse wave is the distance of the crest or the trough from the equilibrium position.

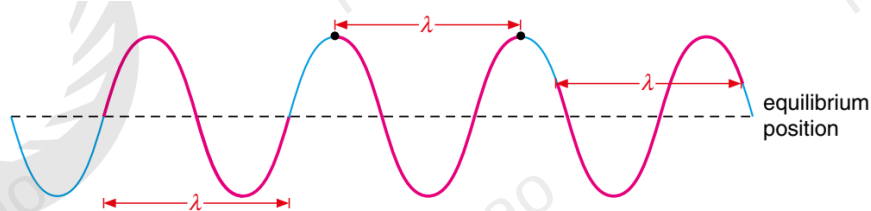


**Fig 4.2a** Amplitude of a transverse wave.

Amplitude is measured from the equilibrium line to a crest or trough; crest-to-trough distance is twice the amplitude.

### 3.4.4

The wavelength of a transverse wave is equal to the distance between two successive crests (or troughs).



**Fig 4.2b** Wavelength of a transverse wave.

One wavelength joins adjacent points in the same phase, such as successive crests. It is a distance along wave propagation.

### 3.4.5

The period of a wave is the time required to produce a complete cycle of the wave. It is also the time for the wave to travel one wavelength.

### 3.4.6

The frequency of a wave is the number of complete cycles of the wave produced in one second.

### 3.4.7

$$f = \frac{1}{T}$$

f: frequency (Hz); T: period (s).

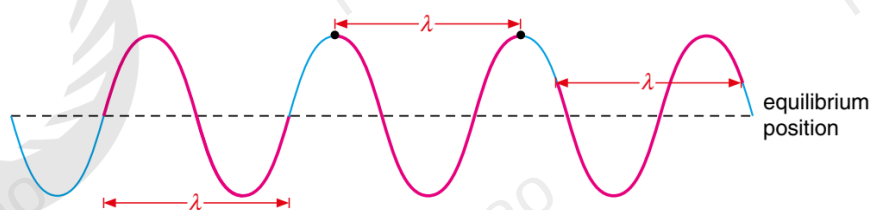
### 3.4.8

The wave speed of a wave is the distance travelled by the wave in one second.

### 3.4.9

$$v = \frac{d}{t} = \frac{\lambda}{T} = f\lambda$$

v: wave speed ( $\text{m s}^{-1}$ ); d: distance travelled by the wave (m); t: time (s);  $\lambda$ : wavelength (m); T: period (s); f: Hz. This is not particle vibration speed.



**Fig 4.2b** Wavelength of a transverse wave.

One wavelength joins adjacent points in the same phase, such as successive crests. It is a distance along wave propagation.

### 3.4.10

A wave travels faster along a stretched spring (or string) when

- 1 the tension  $T$  in the spring increases;
- 2 the mass per unit length  $\mu$  of the spring decreases.

### 3.4.11

For an ideal sinusoidal travelling wave in a uniform medium without attenuation, particles have the same amplitude, period and frequency. Their phases depend on position; particle motion is not the same as propagation of the wave.

## 3B05 Wave Phenomena and Stationary Waves

### 3.5.1

$$\lambda = \frac{v}{f}$$

At fixed wave speed  $v$ , increasing frequency  $f$  decreases wavelength  $\lambda$ .  $v$ :  $\text{m s}^{-1}$ ;  $f$ :  $\text{Hz}$ ;  $\lambda$ :  $\text{m}$ .

### 3.5.2

$$r = i$$

Reflection: incidence angle  $i$  and reflection angle  $r$  are measured from the normal to the reflecting surface.

### 3.5.3

When a water wave crosses from a deep to a shallow region,

- its frequency  $f$  remains unchanged;
- its wavelength  $\lambda$  decreases;
- its wave speed  $v$  decreases.

### 3.5.4

Refraction is the change in travelling direction of a wave due to a change in wave speed.

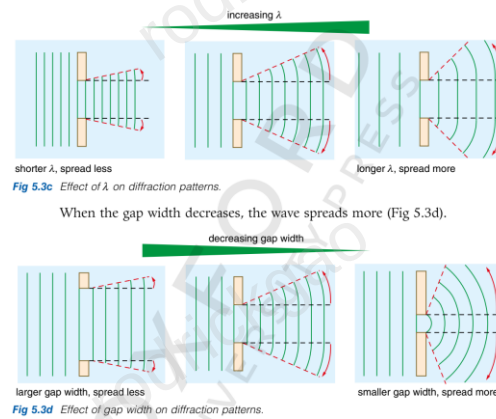
### 3.5.5

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

Refraction: frequency remains unchanged. Angles are from the normal;  $v_1, v_2$ : wave speeds;  $\lambda_1, \lambda_2$ : wavelengths.

### 3.5.6

Diffraction is the bending or spreading of waves around the edge into the 'shadow' of an obstacle.



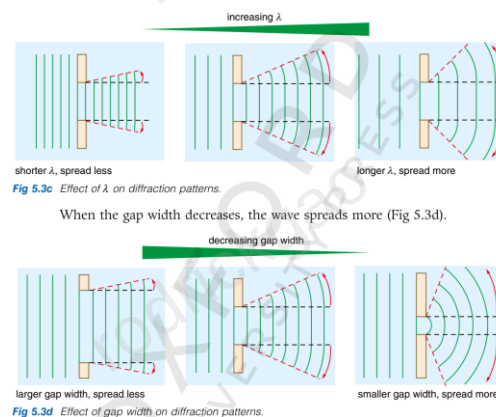
When the gap width decreases, the wave spreads more (Fig 5.3d).

Compare one variable at a time: greater wavelength at fixed gap, or a narrower gap at fixed wavelength, produces greater spreading.

### 3.5.7

When a wave passes through a gap, it spreads more when

- 1 the wavelength increases;
- 2 the gap width decreases.



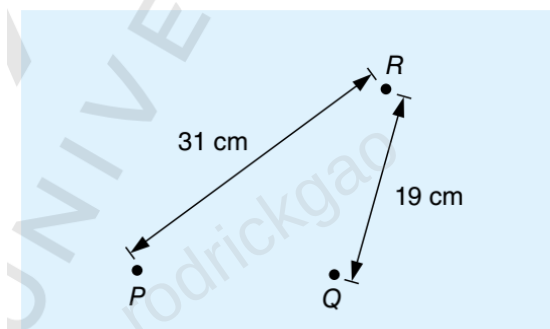
Compare one variable at a time: greater wavelength at fixed gap, or a narrower gap at fixed wavelength, produces greater spreading.

### 3.5.8

When two or more waves overlap, the displacement of any point on the resultant wave is the sum of the displacements due to all waves at that point.

### 3.5.9

In-phase waves produce constructive interference and a larger resultant amplitude. Antiphase waves produce destructive interference; complete cancellation and stationary particles require equal amplitudes at that position.

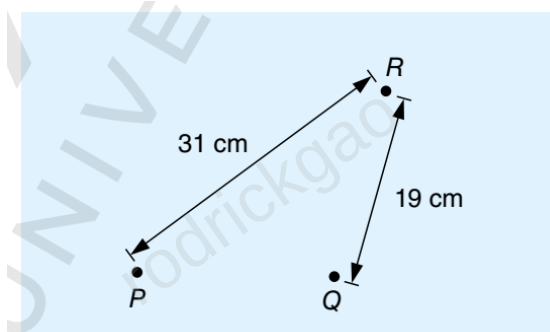


For in-phase sources P and Q, the path difference at R is  $31 - 19 = 12$  cm. At wavelength 4 cm this is 3 wavelengths, giving constructive interference.

### 3.5.10

For two coherent point sources vibrating in phase:

- Constructive interference occurs at positions where the path difference is equal to whole numbers of wavelengths ( $0, \lambda, 2\lambda, \dots$ )
- Destructive interference occurs at positions where the path difference is equal to  $0.5\lambda, 1.5\lambda, 2.5\lambda, \dots$



For in-phase sources P and Q, the path difference at R is  $31 - 19 = 12$  cm. At wavelength 4 cm this is 3 wavelengths, giving constructive interference.

### 3.5.11

The nodal and antinodal lines in the interference pattern become more widely spaced when:

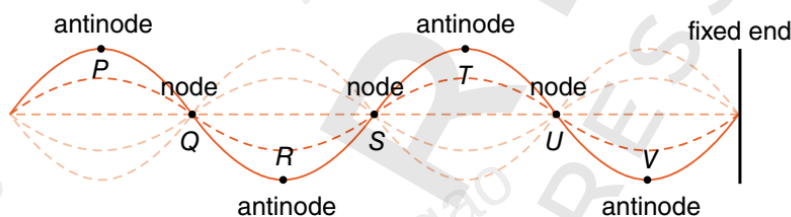
- the wavelength of the waves increases;
- the separation between the sources decreases.

### 3.5.12

All waves exhibit reflection, refraction, diffraction and interference. Among these phenomena, only diffraction and interference are unique to waves.

### 3.5.13

In general, a stationary wave results from the interference of two waves with the same frequency and amplitude travelling at the same speed in opposite directions.



**Fig 5.4f** Nodes and antinodes on a stationary wave.

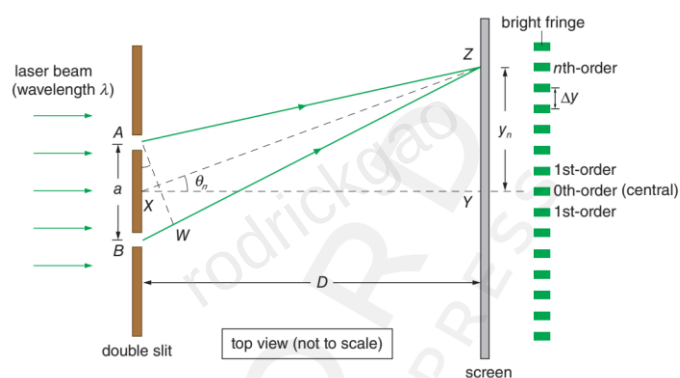
Nodes remain stationary; antinodes have maximum amplitude. Adjacent nodes are half a wavelength apart, and a node-to-adjacent-antinode gap is a quarter wavelength.

## 3B06 Wave Nature of Light

### 3.6.1

In the interference pattern formed by a double slit, the fringe separation  $\Delta y$  increases when:

- 1 the slit separation  $a$  decreases;
- 2 the wavelength  $\lambda$  of light increases;
- 3 the distance  $D$  between the double slit and the screen increases.



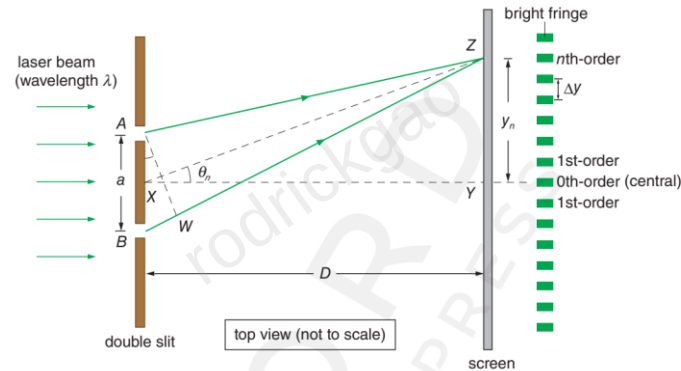
**Fig 6.2i** Investigating the relationship between  $\Delta y$ ,  $\lambda$ ,  $D$  and  $a$

$a$  is slit separation,  $D$  is screen distance and  $\Delta y$  is centre-to-centre fringe spacing. The small-angle approximation is required for  $\Delta y \approx \lambda D/a$ .

### 3.6.2

$$\Delta y \approx \frac{\lambda D}{a}$$

Young's double slit:  $\Delta y$ : adjacent fringe spacing;  $\lambda$ : wavelength;  $D$ : slit-to-screen distance;  $a$ : slit separation. Same length unit;  $D \gg a$  and small angles; coherent illumination.

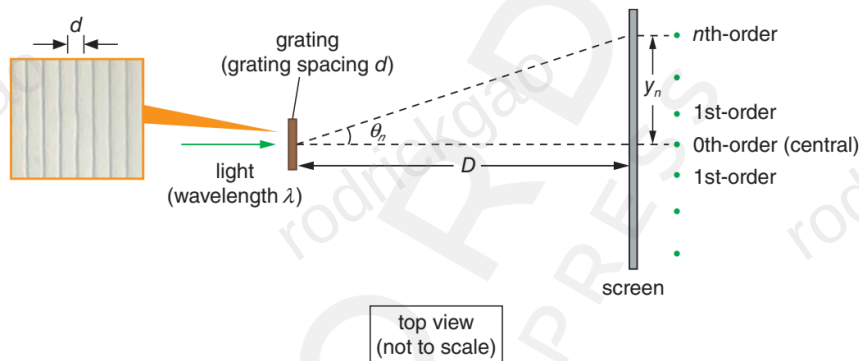


**Fig 6.2i** Investigating the relationship between  $\Delta y$ ,  $\lambda$ ,  $D$  and  $a$

$a$  is slit separation,  $D$  is screen distance and  $\Delta y$  is centre-to-centre fringe spacing. The small-angle approximation is required for  $\Delta y \approx \lambda D/a$ .

### 3.6.3

For a fixed grating order, decreasing grating spacing or increasing wavelength increases the diffraction angle. Increasing screen distance increases the fringe displacement on the screen but does not change its angle. Grating fringes are not generally equally spaced.



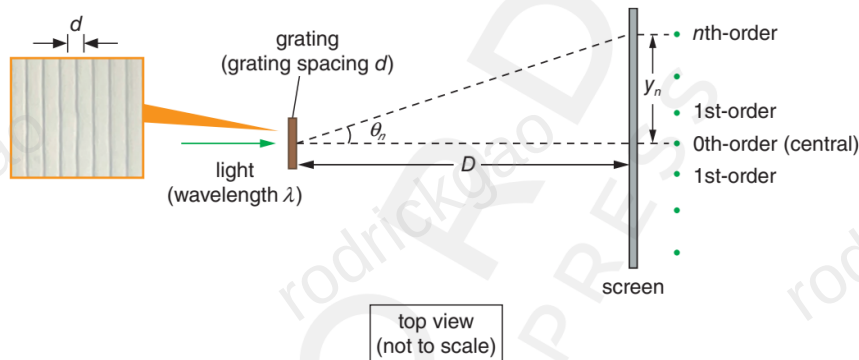
**Fig 6.3i** Fringes formed when light passes through a grating.

Measure  $\theta$  from the central direction.  $d$  is adjacent slit spacing, not the grating width; changing  $D$  changes screen positions but not diffraction angles.

### 3.6.4

$$d \sin \theta = n \lambda$$

Grating with normal incidence:  $d$ : adjacent slit spacing (m);  $\theta$ : angle from the straight-through direction;  $n$ : integer order;  $\lambda$ : wavelength (m).



**Fig 6.3i** Fringes formed when light passes through a grating.

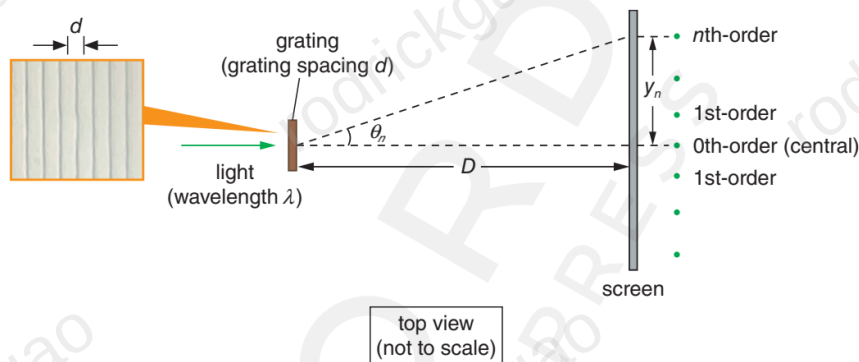
Measure  $\theta$  from the central direction.  $d$  is adjacent slit spacing, not the grating width; changing  $D$  changes screen positions but not diffraction angles.

### 3.6.5

$$n \leq \frac{d}{\lambda}$$

$$n_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor$$

$n$ : non-negative integer order;  $d$ : grating spacing;  $\lambda$ : wavelength, same units. Equality gives  $\theta = 90^\circ$ , a grazing order that does not reach a forward screen.



**Fig 6.3i** Fringes formed when light passes through a grating.

Measure  $\theta$  from the central direction.  $d$  is adjacent slit spacing, not the grating width; changing  $D$  changes screen positions but not diffraction angles.

## 3B07 Sound

### 3.7.1

$$v_{\text{solid}} > v_{\text{liquid}} > v_{\text{gas}}$$

Typical sound-speed ordering for common materials, not a universal inequality for every solid, liquid and gas.  $v$ : sound speed ( $\text{m s}^{-1}$ ).

### 3.7.2

Sound exhibits reflection, refraction, diffraction and interference.

### 3.7.3

$$20\text{Hz} \lesssim f \lesssim 20000\text{Hz}$$

Approximate audible range for a young healthy human; sensitivity and upper limit vary with age and the individual.

### 3.7.4

$$f > 20000\text{Hz}$$

Ultrasound has frequency above approximately 20 kHz, the conventional upper limit of human hearing.

### 3.7.5

The pitch of a note increases with the frequency of the sound wave.

The loudness of a note increases with the amplitude of the sound wave.

Notes of different qualities have different waveforms.

# Compulsory Book 4

## 401 Electrostatics

### 4.1.1

Like charges repel; unlike charges attract.

### 4.1.2

The net charge of an isolated system is conserved.

### 4.1.3

A neutral object is attracted by a charged object.

### 4.1.4

$$F = \frac{|Q_1 Q_2|}{4\pi\epsilon_0 r^2}$$

F: force magnitude (N);  $Q_1, Q_2$ : charges (C); r: separation (m);  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ . Point charges in vacuum/air; like signs repel, opposite signs attract.

### 4.1.5

$$\vec{E} = \frac{\vec{F}}{q_0}$$

E: electric field ( $\text{N C}^{-1}$ ); F: force on a small positive test charge (N);  $q_0$ : test charge (C). A negative charge experiences force opposite to E.

### 4.1.6

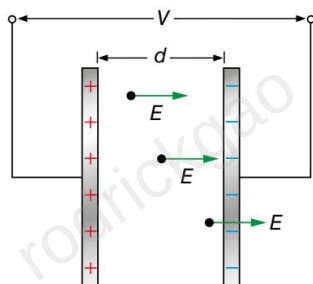
$$E = \frac{|Q|}{4\pi\epsilon_0 r^2}$$

E: field magnitude ( $\text{N C}^{-1}$ ); Q: source charge (C); r: distance (m);  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ . Field points away from +Q and towards -Q.

### 4.1.7

$$E = \frac{V}{d}$$

Uniform field between large parallel plates, away from edges. E: magnitude ( $\text{V m}^{-1}$ ); V: potential difference magnitude (V); d: separation (m). Field points from high to low potential.



**Fig 1.3j** E is the same everywhere between two parallel charged plates.

The field is approximately uniform away from edges and points from the positive plate towards the negative plate. d is the perpendicular plate spacing.

### 4.1.8

$$F = \frac{|Q_1 Q_2|}{4\pi\epsilon_0 r^2}$$

F: force magnitude (N);  $Q_1$ ,  $Q_2$ : charges (C); r: separation (m);  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ . Point charges in vacuum/air; like signs repel, opposite signs attract.

### 4.1.9

$$E = \frac{|Q|}{4\pi\epsilon_0 r^2}$$

E: field magnitude ( $\text{N C}^{-1}$ ); Q: source charge (C); r: distance (m);  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ . Field points away from +Q and towards -Q.

## 402 Current, Voltage and Resistance

### 4.2.1

An electric current is the rate of flow of electric charges.

### 4.2.2

$$I_{\text{av}} = \frac{Q}{\Delta t}$$

$I_{\text{av}}$ : average current (A); Q: charge passing a cross-section (C);  $\Delta t$ : time (s). Equals current for steady flow.

### 4.2.3

By convention, current flows from the positive terminal to the negative terminal of a battery via a circuit.

### 4.2.4

- Current is the same throughout a series branch.
- At a junction, total current entering equals total current leaving, provided charge does not accumulate there.

### 4.2.5

$$\varepsilon = \frac{E_{\text{supplied}}}{Q}$$

$\varepsilon$ : source e.m.f. (V);  $E_{\text{supplied}}$ : energy supplied by the source (J);  $Q$ : charge moved (C). Energy per charge, not a force.

### 4.2.6

$$V = \frac{E_{\text{transferred}}}{Q}$$

$V$ : potential difference (V);  $E_{\text{transferred}}$ : electrical energy converted in the load (J);  $Q$ : charge (C). For a signed potential change use  $\Delta V = \Delta U/q$ .

### 4.2.7

$$V = \frac{E_{\text{transferred}}}{Q}$$

$V$ : potential difference (V);  $E_{\text{transferred}}$ : electrical energy converted in the load (J);  $Q$ : charge (C). For a signed potential change use  $\Delta V = \Delta U/q$ .

### 4.2.8

- The potential differences across series loads add to the applied terminal voltage. This equals the source e.m.f. only if internal voltage loss is negligible.
- Parallel branches connected between the same two nodes have the same potential difference.

### 4.2.9

Series cell e.m.f.s add algebraically according to polarity. Identical cells connected in parallel with matching polarity have the same e.m.f. as one cell. Unequal ideal voltage sources must not be treated as interchangeable parallel cells.

#### 4.2.10

$$R = \frac{V}{I}$$

R: resistance ( $\Omega$ ); V: voltage across component (V); I: current through it (A). Definition at an operating point; constant R additionally requires ohmic behaviour and fixed physical conditions.

#### 4.2.11

$$V \propto I$$

Ohm's law: temperature and other physical conditions remain constant. V: voltage; I: current.

#### 4.2.12

$$R = \frac{\rho l}{A}$$

Uniform wire: R ( $\Omega$ ); resistivity  $\rho$  ( $\Omega \text{ m}$ ); length  $l$  (m); cross-sectional area A ( $\text{m}^2$ ). Temperature fixed.

#### 4.2.13

The resistance of a wire depends on its length, cross-sectional area, resistivity (material) and temperature.

#### 4.2.14

$$R_{\text{eq}} = R_1 + R_2 + \dots + R_n$$

Resistors in series carry the same current.  $R_{\text{eq}}$ : equivalent resistance ( $\Omega$ );  $R_i$ : individual resistances ( $\Omega$ ).

#### 4.2.15

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

Exactly two resistors in parallel, with the same voltage across both; all resistances in  $\Omega$ .

#### 4.2.16

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

Parallel branches share the same voltage.  $R_{\text{eq}}$  and  $R_i$  in  $\Omega$ ; reciprocal resistances add.

#### 4.2.17

$$P_{av} = \frac{E}{\Delta t}$$

$P_{av}$ : average power (W);  $E$ : energy transferred (J);  $\Delta t$ : time (s).

#### 4.2.18

$$P = VI$$

$P$ : electrical power (W);  $V$ : p.d. (V);  $I$ : current (A). Steady DC values or simultaneous instantaneous values.

#### 4.2.19

$$P = VI = I^2 R = \frac{V^2}{R}$$

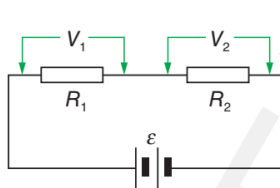
Resistive load;  $P$ : W;  $V$ : V;  $I$ : A;  $R$ :  $\Omega$ . Use DC values, or RMS voltage/current for a constant resistor in AC.

### 403 Practical Circuits

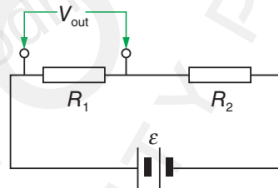
#### 4.3.1

$$V_{out} = \frac{R_1}{R_1 + R_2} \varepsilon$$

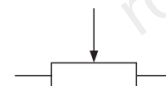
Unloaded potential divider. Output is across  $R_1$  in the source diagram.  $R_1$ ,  $R_2$ :  $\Omega$ ;  $\varepsilon$ ,  $V_{out}$ : V. A connected load changes the effective resistance.



**Fig 3.1d** Resistors  $R_1$  and  $R_2$  connected in series to a battery.



**Fig 3.1e** The output voltage depends on the resistance ratio.



**Fig 3.1f** Circuit symbol of potential divider.

The output terminals are across  $R_1$ . This is why  $R_1$ , rather than  $R_2$ , appears in the numerator of the displayed divider formula.

#### 4.3.2

A real ammeter is connected in series. Its internal resistance can reduce the current being measured, so a good ammeter has very low resistance; an ideal ammeter has zero resistance.

### 4.3.3

A voltmeter is connected in parallel. It draws some current and can reduce the measured voltage through loading of a circuit with nonzero source resistance. A good voltmeter has very high resistance; an ideal voltmeter has infinite resistance.

### 4.3.4

$$\varepsilon = V + Ir$$

Discharging cell:  $\varepsilon$ : e.m.f. (V);  $V$ : terminal voltage (V);  $I$ : supplied current (A);  $r$ : internal resistance ( $\Omega$ ). During charging the terminal-voltage relation changes sign.

## 404 Mains Electricity

### 4.4.1

The potential of the neutral wire stays at 0 V.

The potential of the live wire keeps changing between a positive peak value and a negative peak value.

### 4.4.2

Safety designs in mains circuit:

1 Fuse and circuit breaker

- break the circuit when current is too large
- prevent circuit from overheating
- rating slightly larger than normal operating current
- installed in live wire

2 Earth wire

- provide a path for current to flow when a fault develops
- prevent electric shock

3 Double insulation (plastic case)

- prevent electric shock

### 4.4.3

An alternating current (or voltage) changes direction periodically while a direct current does not change direction.

#### 4.4.4

$$P = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} = V_{\text{rms}} I_{\text{rms}}$$

Pure constant resistance  $R$  ( $\Omega$ ), so voltage and current are in phase. Average power  $\bar{P}$ : W; RMS voltage: V; RMS current: A.

#### 4.4.5

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$$
$$I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}}$$

Only for sinusoidal voltage/current with zero DC offset. Peak means amplitude, not peak-to-peak.

### 405 Magnetic Fields

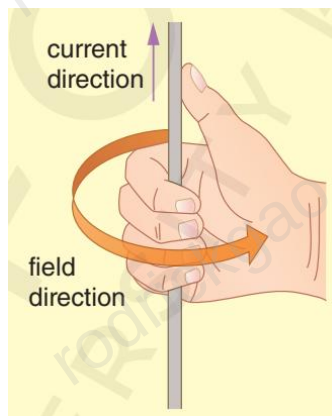
#### 4.5.1

Like poles repel; unlike poles attract.

#### 4.5.2

The magnetic field pattern around a long straight wire:

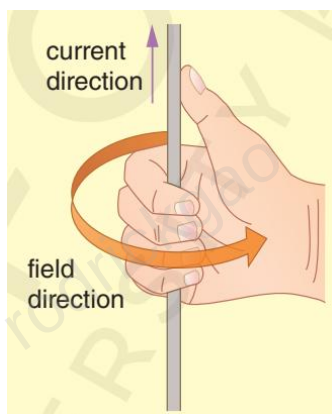
- 1 The field lines form circles with the wire as the centre.
- 2 Reversing the current reverses the direction of the field lines.
- 3 The field lines are more closely-spaced (the magnetic field is stronger) in the region closer to the wire.



Right thumb follows conventional current; curled fingers follow the circular magnetic field. Reverse the current and the field direction reverses.

### 4.5.3

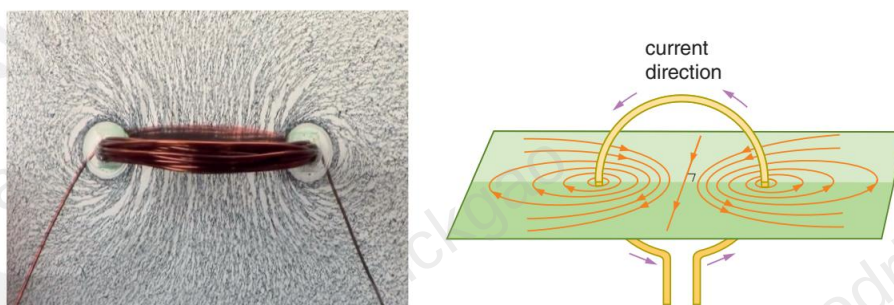
Right-hand grip rule for a straight current-carrying wire: point your right thumb in the direction of conventional current. The curl of your fingers gives the direction of the circular magnetic field lines around the wire.



Right thumb follows conventional current; curled fingers follow the circular magnetic field. Reverse the current and the field direction reverses.

### 4.5.4

At the centre of a circular coil, the field line is straight and at a right angle to the plane of the coil; other field lines run in loops.



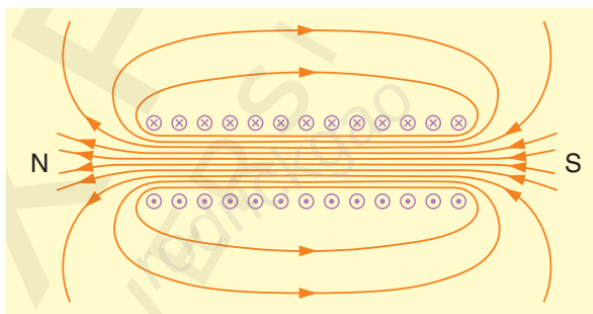
**Fig 5.2e** Magnetic field pattern around a current-carrying circular coil.

The coil field is perpendicular to the coil plane at its centre. Field lines form closed loops around the current.

#### 4.5.5

Inside a long solenoid, the field lines are straight, parallel and evenly spaced, so the magnetic field is approximately uniform.

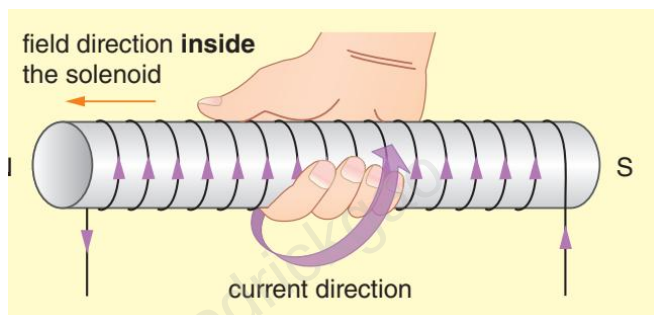
Outside the solenoid, the field pattern is similar to that of a bar magnet: one end is a north pole and the other is a south pole.



Inside the long solenoid the nearly parallel, evenly spaced field lines indicate an approximately uniform field. Outside, they run from N towards S.

#### 4.5.6

Right-hand grip rule for a solenoid: curl the fingers of your right hand in the direction of conventional current around the turns. Your extended thumb points along the magnetic field inside the solenoid and towards its north pole.

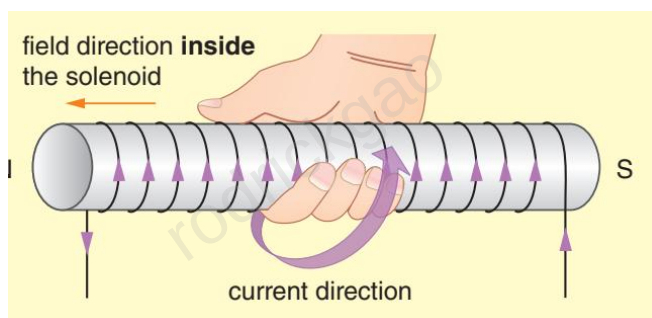


Curl the fingers along conventional current around the turns. The thumb points along the internal magnetic field and towards the north end.

### 4.5.7

An electromagnet can be made stronger by:

- 1 increasing the number of turns per unit length of the coil;
- 2 increasing the current through the coil;
- 3 inserting a soft-iron core into the coil.



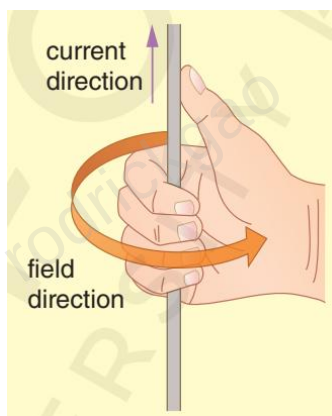
Curl the fingers along conventional current around the turns. The thumb points along the internal magnetic field and towards the north end.

### 4.5.8

$$B = \frac{\mu_0 I}{2\pi r}$$

Long straight wire in vacuum/air. B: T; I: A; perpendicular distance r: m;  $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ .

Direction: right-hand grip rule.



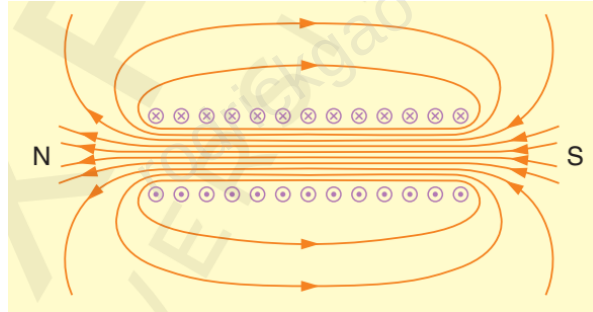
Right thumb follows conventional current; curled fingers follow the circular magnetic field. Reverse the current and the field direction reverses.

#### 4.5.9

$$B = \frac{\mu_0 NI}{l} = \mu_0 nI$$

$$n = \frac{N}{l}$$

Inside a long air-core solenoid, away from ends. B: T; N: turns; l: length (m); n: turns per metre; I: A;  
 $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ .

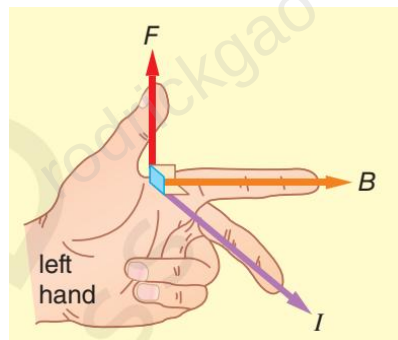


Inside the long solenoid the nearly parallel, evenly spaced field lines indicate an approximately uniform field. Outside, they run from N towards S.

### 406 Force Due to a Magnetic Field

#### 4.6.1

Fleming's left-hand rule: hold the thumb, first finger and second finger mutually perpendicular. First finger: external magnetic field B; second finger: conventional current I; thumb: magnetic force F. For a moving negative charge, reverse the direction obtained using its velocity as the current direction.



First finger B, second finger I and thumb F are mutually perpendicular. I means conventional current; a negative moving charge reverses the force obtained from its velocity.

#### 4.6.2

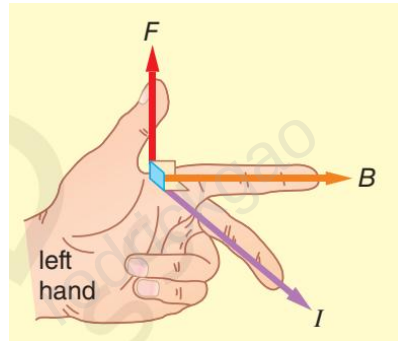
$$1\text{T} = 1\text{N A}^{-1}\text{m}^{-1}$$

Tesla is magnetic flux density's SI unit; from  $F = BIl$  with the wire perpendicular to the field.

### 4.6.3

$$F = BIl\sin\theta$$

F: force magnitude (N); B: magnetic flux density (T); I: conventional current (A); l: wire length in field (m);  $\theta$ : angle between current direction and B. Uniform field and straight wire.



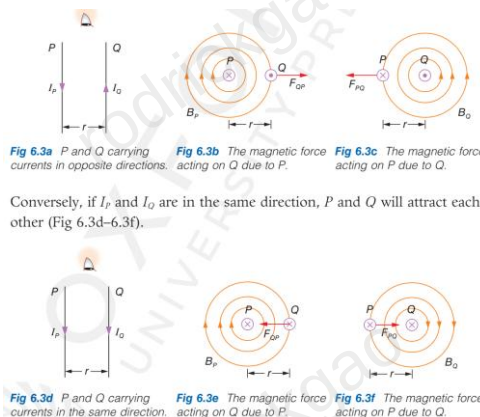
First finger B, second finger I and thumb F are mutually perpendicular. I means conventional current; a negative moving charge reverses the force obtained from its velocity.

### 4.6.4

In a simple DC motor, the current-carrying coil experiences magnetic forces that turn it. The split-ring commutator reverses the coil current every half-turn so the turning effect continues in the same rotational direction.

### 4.6.5

Two wires repel each other if they carry currents in opposite directions. They attract each other if they carry currents in the same direction (Fig 6.3g).

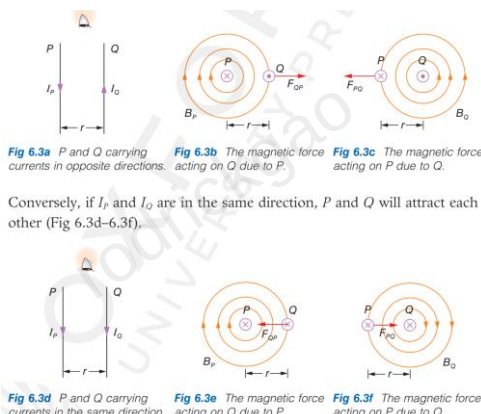


Use the field of one wire to find the force on the other. Opposite currents repel (top); currents in the same direction attract (bottom).

#### 4.6.6

$$\frac{F}{l} = \frac{\mu_0 I_P I_Q}{2\pi r}$$

Parallel long wires: force per length (N m<sup>-1</sup>); current magnitudes I<sub>P</sub>, I<sub>Q</sub> (A); separation r (m). Same current direction: attraction; opposite: repulsion.

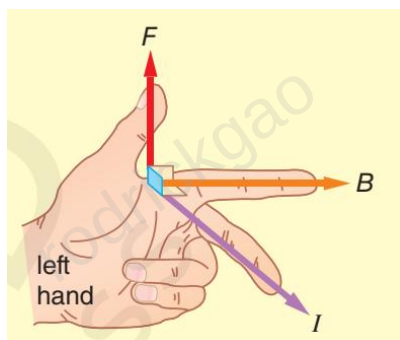


Use the field of one wire to find the force on the other. Opposite currents repel (top); currents in the same direction attract (bottom).

#### 4.6.7

$$F = |Q|vB\sin\theta$$

F: magnetic-force magnitude (N); Q: charge (C); v: speed (m s<sup>-1</sup>); B: T; θ: angle between velocity and field. Direction reverses for negative charge; the magnetic force does no work.



First finger B, second finger I and thumb F are mutually perpendicular. I means conventional current; a negative moving charge reverses the force obtained from its velocity.

## 407 Electromagnetic Induction

### 4.7.1

An e.m.f. is induced when magnetic flux linkage changes. Its magnitude depends on the rate of change of flux linkage. Faster motion, a stronger field or more turns can increase it when they increase that rate. A closed conducting path is also needed for an induced current.

#### 4.7.2

An e.m.f. is induced when magnetic flux linkage changes. Its magnitude depends on the rate of change of flux linkage. Faster motion, a stronger field or more turns can increase it when they increase that rate. A closed conducting path is also needed for an induced current.

#### 4.7.3

The induced current (or e.m.f.) is in the direction such that it opposes (or tends to oppose) the change causing it.

#### 4.7.4

In a simple generator, when the coil is spun, the two sides of the coil cut the magnetic field lines. This induces an e.m.f. in the coil and drives a current. The current flowing in the coil reverses its direction every half turn.

- In a d.c. generator, the commutator reverses the connections of the coil with the external circuit every half turn. Hence, a d.c. flows in the external circuit.
- In an a.c. generator, the current flows through the slip rings to the external circuit. Hence, an a.c. flows in the external circuit.

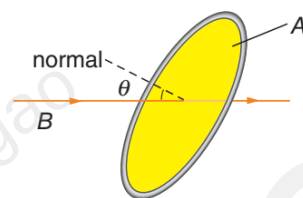
#### 4.7.5

Changing magnetic flux through paths inside a bulk conductor induces circulating eddy currents. Their magnetic effect opposes the change causing them and may produce heating or magnetic braking.

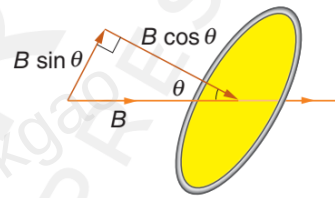
#### 4.7.6

$$\Phi = BA \cos \theta$$

Uniform field through a flat area:  $\Phi$  (Wb);  $B$  (T);  $A$  ( $\text{m}^2$ );  $\theta$ : angle between  $B$  and the surface NORMAL, not the plane.



**Fig 7.3c**  $B$  making an angle  $\theta$  with the normal of the coil.



**Fig 7.3d** Components of  $B$  parallel and perpendicular to the normal of the coil.

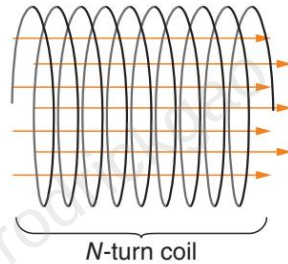
$\theta$  is between  $B$  and the area normal. Only  $B \cos \theta$  passes normally through the area; a field parallel to the surface gives zero flux.

#### 4.7.7

$$\Psi = N\Phi$$

$\Psi$ : magnetic flux linkage (Wb turn);  $N$ : turns;  $\Phi$ : flux through each turn (Wb). All turns link the same flux.

magnetic flux linkage =  $N\Phi$



**Fig 7.3e** Magnetic flux linkage through an  $N$ -turn coil.

Each of  $N$  turns links flux  $\Phi$ , so total flux linkage is  $N\Phi$ . A change in this total determines the induced e.m.f.

#### 4.7.8

$$1\text{T} = 1\text{Wbm}^{-2}$$

Magnetic flux density equals flux per perpendicular area for a uniform field.

#### 4.7.9

Faraday's law: the magnitude of induced e.m.f. equals the rate of change of magnetic flux linkage. For one turn, flux linkage is the flux through that turn. Lenz's law supplies the direction/sign.

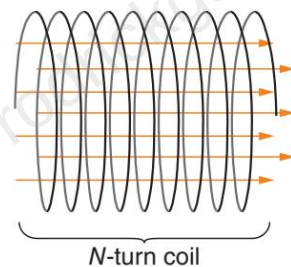
#### 4.7.10

$$\varepsilon_{\text{av}} = -\frac{\Delta\Phi}{\Delta t} (N = 1)$$

$$\varepsilon_{\text{av}} = -N \frac{\Delta\Phi}{\Delta t}$$

$\varepsilon$ : average induced e.m.f. (V);  $\Phi$ : flux per turn (Wb);  $N$ : fixed turns;  $\Delta t$ : s. Minus sign expresses Lenz's law. Instantaneous e.m.f. uses the instantaneous rate of change.

magnetic flux linkage =  $N\Phi$



**Fig 7.3e** Magnetic flux linkage through an  $N$ -turn coil.

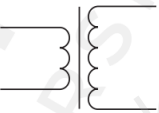
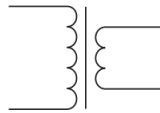
Each of  $N$  turns links flux  $\Phi$ , so total flux linkage is  $N\Phi$ . A change in this total determines the induced e.m.f.

## 408 Transformers and Transmission of Electricity

### 4.8.1

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

Ideal transformer: primary/secondary voltages  $V_p$ ,  $V_s$  (V), and turns  $N_p$ ,  $N_s$ . Alternating flux; use matching RMS or instantaneous values.

|                 | Step-up transformer                                                                 | Step-down transformer                                                                |
|-----------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Voltage         | $V_p < V_s$                                                                         | $V_p > V_s$                                                                          |
| Number of turns | $N_p < N_s$                                                                         | $N_p > N_s$                                                                          |
| Symbol          |  |  |

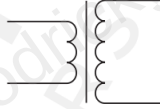
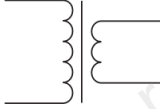
**Table 8.1a** Step-up transformer and step-down transformer.

Step-up: more secondary turns and a higher secondary voltage. Step-down: fewer secondary turns and a lower voltage; ideal power remains conserved.

#### 4.8.2

$$V_p I_p = V_s I_s$$

Ideal transformer with no power loss and resistive load; p: primary; s: secondary; voltage in V and current in A.

|                 | Step-up transformer                                                               | Step-down transformer                                                               |
|-----------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Voltage         | $V_p < V_s$                                                                       | $V_p > V_s$                                                                         |
| Number of turns | $N_p < N_s$                                                                       | $N_p > N_s$                                                                         |
| Symbol          |  |  |

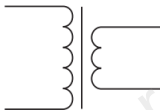
**Table 8.1a** Step-up transformer and step-down transformer.

Step-up: more secondary turns and a higher secondary voltage. Step-down: fewer secondary turns and a lower voltage; ideal power remains conserved.

#### 4.8.3

$$\eta V_p I_p = V_s I_s$$

Efficiency  $\eta$  = output/input power, written as a fraction from 0 to 1. Divide a percentage by 100 before substitution. Resistive load; RMS AC values.

|                 | Step-up transformer                                                                 | Step-down transformer                                                                 |
|-----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Voltage         | $V_p < V_s$                                                                         | $V_p > V_s$                                                                           |
| Number of turns | $N_p < N_s$                                                                         | $N_p > N_s$                                                                           |
| Symbol          |  |  |

**Table 8.1a** Step-up transformer and step-down transformer.

Step-up: more secondary turns and a higher secondary voltage. Step-down: fewer secondary turns and a lower voltage; ideal power remains conserved.

#### 4.8.4

The power loss in transmission cables can be reduced if electricity is transmitted at a high voltage.

# Compulsory Book 5

## 501 Radiation and Radioactivity

### 5.1.1

X-rays are EM waves of short wavelengths. They are produced by hitting heavy metals with fast electrons. They are ionizing and dangerous. They carry high energy and have high penetrating power, and can be used to reveal internal structures of objects.

### 5.1.2

Nuclear radiation is the high-energy radiation emitted spontaneously by unstable nuclei. The three common types of nuclear radiation are  $\alpha$ ,  $\beta$  and  $\gamma$ .

### 5.1.3

Ionizing power of the nuclear radiation:

$$\alpha > \beta > \gamma$$

### 5.1.4

Typical ranges in air: alpha particles travel a few centimetres and beta particles a few metres. Gamma rays are attenuated gradually and may travel over 100 m; unlike charged particles, they have no sharp finite range. Values depend on radiation energy and medium.

### 5.1.5

$$C_{\text{net}} = C_{\text{measured}} - C_{\text{background}}$$

Corrected count rate subtracts background measured in the same units. Count rate is not automatically the activity because detector efficiency and geometry matter.

### 5.1.6

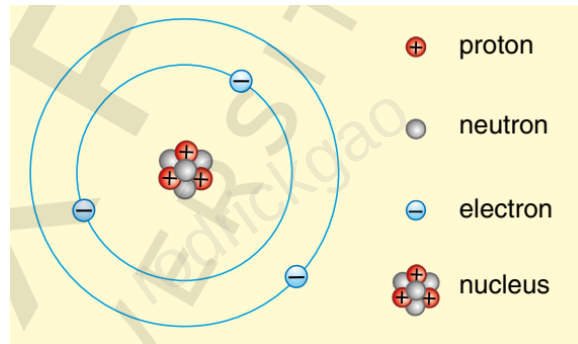
Penetrating power of the nuclear radiation:

$$\gamma > \beta > \alpha$$

## 502 Radioactive Decay

### 5.2.1

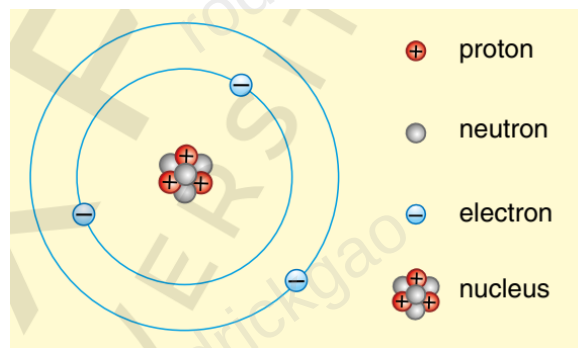
An atom consists of a tiny nucleus containing protons and neutrons, with electrons surrounding it. Most of the atom is empty space.



Protons and neutrons form the nucleus; electrons occupy the surrounding region. The diagram is schematic and not to scale.

### 5.2.2

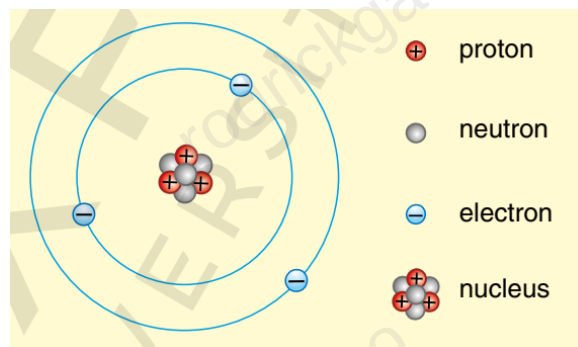
The atomic number ( $Z$ ) is the number of protons in the nucleus of an atom. The mass number ( $A$ ) is the total number of protons and neutrons in the nucleus of an atom.



Protons and neutrons form the nucleus; electrons occupy the surrounding region. The diagram is schematic and not to scale.

### 5.2.3

Isotopes are different atoms that contain the same number of protons but different numbers of neutrons.



Protons and neutrons form the nucleus; electrons occupy the surrounding region. The diagram is schematic and not to scale.

#### 5.2.4

$$A = kN$$

A: activity ( $\text{Bq} = \text{s}^{-1}$ ); k: decay constant ( $\text{s}^{-1}$ ); N: undecayed nuclei. Activity is positive; the decay rate of N is negative.

#### 5.2.5

The half-life of a radioactive substance is the time over which the number of radioactive nuclei decreases by half. It is also equal to the time taken for the activity of a radioactive sample to fall by half.

#### 5.2.6

$$N = N_0 \left(\frac{1}{2}\right)^n$$

$$A = A_0 \left(\frac{1}{2}\right)^n$$

$$n = \frac{t}{t_{1/2}}$$

n: elapsed number of half-lives, not necessarily an integer; N: undecayed nuclei; A: activity; subscript 0: initial value. No replenishment; single radionuclide.

#### 5.2.7

$$N = N_0 e^{-kt}$$

$$A = A_0 e^{-kt}$$

k: decay constant ( $\text{s}^{-1}$ ); t: elapsed time (s); N: undecayed nuclei; A: activity (Bq); subscript 0: initial value. Single radionuclide with no replenishment.

#### 5.2.8

$$t_{1/2} = \frac{\ln 2}{k}$$

$t_{1/2}$ : half-life (s); k: decay constant ( $\text{s}^{-1}$ ); ln: natural logarithm. Use matching time units.

### 503 Nuclear Energy

#### 5.3.1

Nuclear fission is a nuclear reaction in which a nucleus splits into lighter nuclei.

### 5.3.2

A chain reaction is sustained if at least one neutron on average from each fission causes another fission. The minimum amount of fissile material required for a sustained chain reaction is called the critical mass.

### 5.3.3

In nuclear fusion, two nuclei join together to form a heavier nucleus. A high temperature is required to start a fusion reaction.

### 5.3.4

$$E_{\text{released}} = \Delta mc^2$$

$\Delta m$  = mass before – mass after (kg), positive for an energy-releasing reaction; E: J;  $c \approx 3.00 \times 10^8 \text{ m s}^{-1}$ .

### 5.3.5

$$1\text{eV} \approx 1.60 \times 10^{-19}\text{J}$$

One electronvolt is the kinetic energy gained by an electron accelerated through a potential difference of 1 V. It is an energy unit, not a voltage unit. Rounded textbook conversion.

### 5.3.6

$$1\text{u} = \frac{m(^{12}\text{C})}{12} \approx 1.661 \times 10^{-27}\text{kg}$$

Definition uses a free neutral carbon-12 atom at rest in its ground state; u is the unified atomic mass unit.

### 5.3.7

$$1\text{u} \approx 1.661 \times 10^{-27}\text{kg}$$

$$(1\text{u})c^2 \approx 931\text{MeV}$$

Convert nuclear mass from u to kg before using  $E = mc^2$  in joules. A mass of 1 u corresponds to rest energy of about 931 MeV, or  $1 \text{ u} \approx 931 \text{ MeV}/c^2$ . Textbook rounded values.

# Elective Book 1

## E101 Introduction to Astronomy

### E1.1.1

$$1\text{AU} \approx 1.50 \times 10^{11}\text{m}$$

One astronomical unit is approximately the mean Earth–Sun distance. Textbook rounded conversion for Solar System distances.

### E1.1.2

One light year (ly) is the distance travelled by light in a vacuum in one year.

### E1.1.3

$$1\text{ly} \approx 9.46 \times 10^{15}\text{m}$$

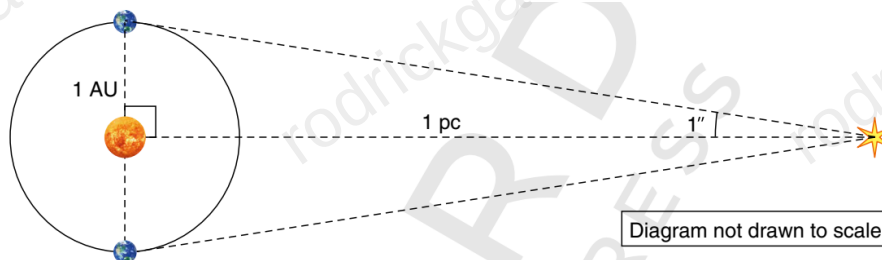
A light-year is the distance travelled by light in vacuum in one year; a unit of distance, not time.

### E1.1.4

$$1\text{pc} \approx 3.09 \times 10^{16}\text{m} \approx 3.26\text{ly}$$

$$1\text{pc} \approx 206265\text{AU}$$

Parsec: distance corresponding to annual parallax of one arcsecond. Rounded conversions.



**Fig 3.1i** Definition of one parsec.

The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

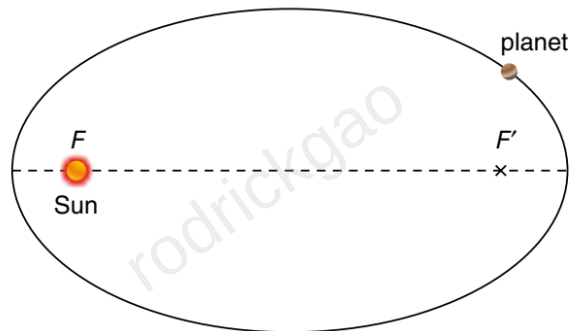
### E1.1.5

Retrograde motion is the apparent westward motion of a planet as seen from the Earth against the background stars.

## E102 Motion of Celestial Bodies under Gravitation

### E1.2.1

Kepler's first law: planets follow elliptical orbits with the Sun at ONE focus, not at the centre or at both foci.



**Fig 2.1a** Kepler's first law.

The Sun lies at one focus F. The other focus is empty; the orbit is not centred on the Sun.

### E1.2.2

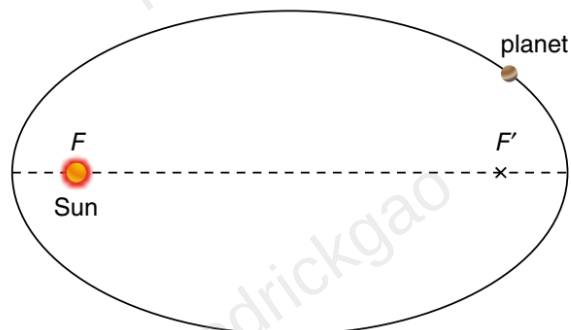
A line from the Sun to any given planet sweeps out equal areas in equal time intervals.

### E1.2.3

$$T^2 \propto a^3$$

$$\frac{T^2}{a^3} = \text{constant}$$

Kepler's third law for bodies orbiting the same dominant central mass. T: period; a: semi-major axis, not instantaneous distance.



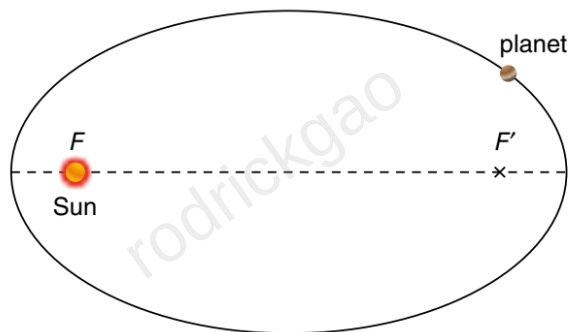
**Fig 2.1a** Kepler's first law.

The Sun lies at one focus F. The other focus is empty; the orbit is not centred on the Sun.

#### E1.2.4

$$T^2 = \frac{4\pi^2}{GM} a^3$$

Orbiting mass much smaller than central mass  $M$  (kg);  $T$ : s;  $a$ : semi-major axis (m);  $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ .



**Fig 2.1a** Kepler's first law.

The Sun lies at one focus  $F$ . The other focus is empty; the orbit is not centred on the Sun.

#### E1.2.5

An object experiences apparent weightlessness if the object and the thing that supports it fall with the same acceleration due to gravity.

#### E1.2.6

$$U = -\frac{GMm}{r}$$

$U$ : gravitational potential energy (J), zero at infinite separation;  $M$ ,  $m$ : kg;  $r$ : centre separation (m). Point masses or non-overlapping spherical masses. Negative  $U$  alone does not imply a bound orbit; total energy must be negative.

#### E1.2.7

$$K_1 + U_1 = K_2 + U_2$$

Only gravity does work;  $K$ : kinetic energy (J);  $U$ : gravitational potential energy (J), with the same zero level in both states.

#### E1.2.8

$$v_e = \sqrt{\frac{2GM}{R}}$$

Escape speed  $v_e$  ( $\text{m s}^{-1}$ );  $M$ : central mass (kg);  $R$ : starting radius from its centre (m). No air resistance or subsequent propulsion; final speed tends to zero at infinity.

## E103 Stars and the Universe

### E1.3.1

$$1^\circ = 60' = 3600''$$

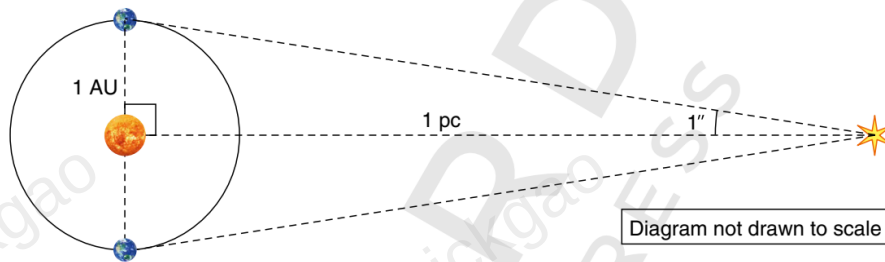
$$1' = 60''$$

Angular units: degree, arcminute and arcsecond. These are angles, not units of time.

### E1.3.2

$$1\text{pc} \approx 206265\text{AU}$$

Small-angle parallax geometry: one radian is approximately 206 265 arcseconds.



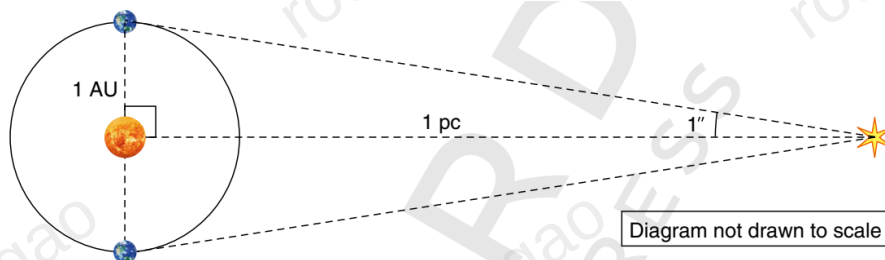
**Fig 3.1i** Definition of one parsec.

The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

### E1.3.3

$$\frac{d}{\text{pc}} = \frac{1}{p/\text{arcsec}}$$

d: distance in parsecs; p: annual parallax angle in arcseconds, half the total angular shift over six months. This numerical relation requires these units.



**Fig 3.1i** Definition of one parsec.

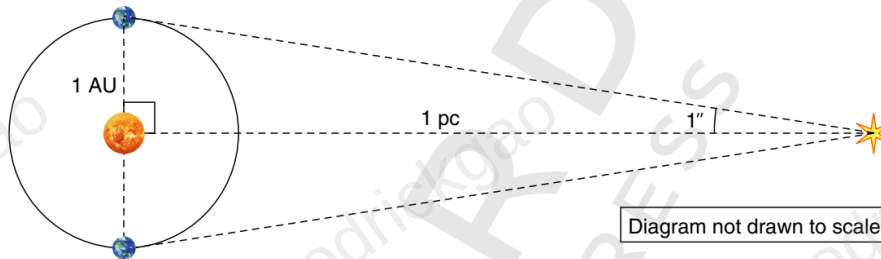
The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

### E1.3.4

$$1\text{pc} \approx 3.09 \times 10^{16}\text{m} \approx 3.26\text{ly}$$

$$1\text{pc} \approx 206265\text{AU}$$

Parsec: distance corresponding to annual parallax of one arcsecond. Rounded conversions.



**Fig 3.1i** Definition of one parsec.

The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

### E1.3.5

$$L = \frac{E_{\text{emitted}}}{\Delta t}$$

Luminosity  $L$  (W): total radiated energy (J) per time (s), emitted in all directions.

### E1.3.6

$$b = \frac{L}{4\pi d^2}$$

Apparent brightness  $b$  ( $\text{W m}^{-2}$ ); luminosity  $L$  (W); distance  $d$  (m). Isotropic emission and negligible absorption.

### E1.3.7

The apparent magnitude  $m$  measures the brightness of celestial bodies as seen by an observer on Earth.

### E1.3.8

the apparent magnitude  $m$  of a star depends on its luminosity and its distance from the Earth.

### E1.3.9

The absolute magnitude  $M$  of a star is defined as the apparent magnitude of the star if it were located at a distance of 10 pc from the Earth.

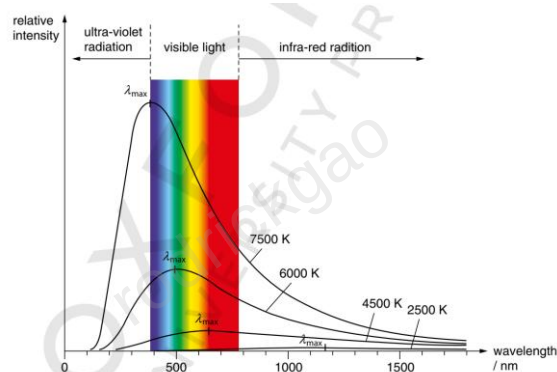
### E1.3.10

the absolute magnitude  $M$  depends only on a star's luminosity and is independent of its distance from the Earth.

### E1.3.11

$$T \uparrow \Rightarrow \lambda_{\max} \downarrow$$

Blackbody spectrum: increasing absolute temperature  $T$  shifts the wavelength of peak emission  $\lambda_{\max}$  towards shorter wavelengths.

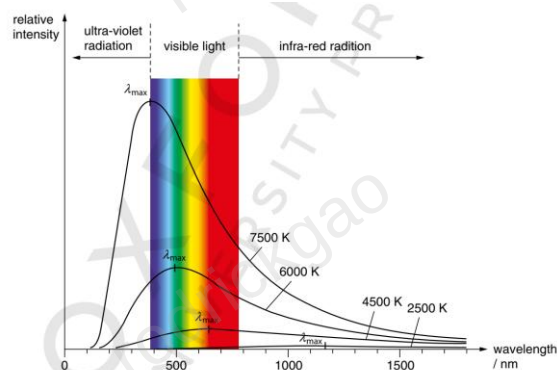


**Fig 3.3d** Blackbody radiation curves at different surface temperatures. A visible spectrum is added to indicate the wavelengths of visible light.

A hotter blackbody has greater total emission and a shorter peak wavelength. The peak position is not itself the perceived colour of the complete spectrum.

### E1.3.12

A cooler star appears redder and a hotter star appears bluer.



**Fig 3.3d** Blackbody radiation curves at different surface temperatures. A visible spectrum is added to indicate the wavelengths of visible light.

A hotter blackbody has greater total emission and a shorter peak wavelength. The peak position is not itself the perceived colour of the complete spectrum.

### E1.3.13

$$J = \sigma T^4$$

J: total radiant power per surface area ( $\text{W m}^{-2}$ ); T: absolute temperature (K);  $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ . Blackbody emission, not net exchange with surroundings.

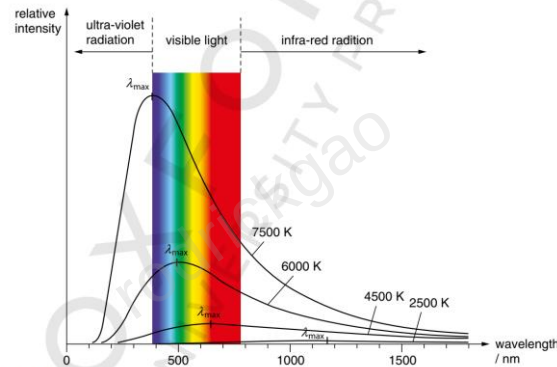


Fig 3.3d Blackbody radiation curves at different surface temperatures. A visible spectrum is added to indicate the wavelengths of visible light.

A hotter blackbody has greater total emission and a shorter peak wavelength. The peak position is not itself the perceived colour of the complete spectrum.

### E1.3.14

$$L = 4\pi R^2 \sigma T^4$$

Spherical blackbody: L: luminosity (W); R: stellar radius (m); T: surface temperature (K);  $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ .

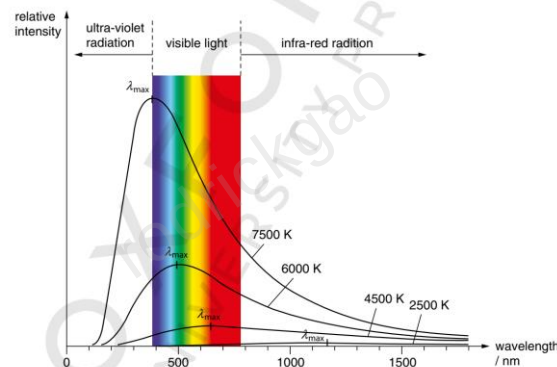


Fig 3.3d Blackbody radiation curves at different surface temperatures. A visible spectrum is added to indicate the wavelengths of visible light.

A hotter blackbody has greater total emission and a shorter peak wavelength. The peak position is not itself the perceived colour of the complete spectrum.

### E1.3.15

$$\frac{L}{L_{\odot}} = \left( \frac{R}{R_{\odot}} \right)^2 \left( \frac{T}{T_{\odot}} \right)^4$$

Compare spherical blackbody stars with the Sun ( $\odot$ ). L: luminosity; R: radius; T: kelvin temperature. Corresponding quantities use identical units.

**E1.3.16**

$$\frac{\Delta\lambda}{\lambda_0} = \frac{\lambda - \lambda_0}{\lambda_0} \approx \frac{v_r}{c}$$

Non-relativistic radial Doppler shift: radial speed is much smaller than  $c$ .  $\lambda_0$ : rest wavelength;  $\lambda$ : observed wavelength;  $v_r > 0$ : receding (redshift);  $v_r < 0$ : approaching.

**E1.3.17**

The greater the distance  $d$  from the Earth, the higher the receding velocity  $v$  of a galaxy, i.e.  $d \uparrow \Rightarrow v \uparrow$ .

# Elective Book 3

## E301 Energy at Home

### E3.1.1

$$\eta_{\%} = \frac{E_{\text{useful}}}{E_{\text{input}}} \times 100\%$$

Percentage efficiency; useful output energy and total input energy in identical units. A fractional efficiency omits  $\times 100\%$ .

### E3.1.2

- Electric hotplate: current in a resistance coil produces heat.
- Induction cooker: an alternating current produces a changing magnetic field, inducing eddy currents in the metal pot; the pot is heated by their heating effect.
- Microwave oven: an oscillating electric field makes polar water molecules rotate rapidly; collisions increase the molecules' total kinetic energy and heat the food.

### E3.1.3

A heat pump is a device that transfers heat against its natural direction of flow.

Ts may tell Ss that the energy conversions of an air conditioner are as shown in the figure below. However, to solve the questions in this book, we usually need to take the air conditioner's electrical power input and work done to be equal. Ss may get confused that why the two quantities are equal even there is energy loss. Ts may tell Ss that this is just an assumption for calculation.

electrical energy

air conditioner

KE of refrigerant, fan and air  
internal energy of refrigerant and compressor  
sound energy

In other words, the amount of heat released to the outside is greater than the heat removed from the room, i.e.  $Q_H > Q_C$ .

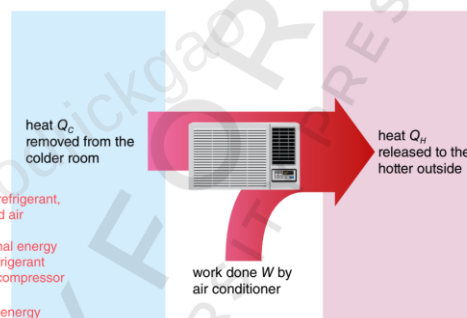


Fig 1.3e  $Q_H = Q_C + W$ .

The heat delivered outside equals heat removed from the cold space plus electrical work input:  $Q_H = Q_C + W$ . The original teacher annotations are retained at the left.

### E3.1.4

$$Q_H = Q_C + W$$

Refrigerator/heat pump per cycle:  $Q_H$ : heat delivered to hot reservoir;  $Q_C$ : heat removed from cold reservoir;  $W$ : work input. All are positive energy magnitudes (J).

⚡ Ts may tell Ss that the energy conversions of an air conditioner are as shown in the figure below. However, to solve the questions in this book, we usually need to take the air conditioner's electrical power input and work done to be equal. Ss may get confused that why the two quantities are equal even there is energy loss. Ts may tell Ss that this is just an assumption for calculation.

In other words, the amount of heat released to the outside is greater than the heat removed from the room, i.e.  $Q_H > Q_C$ .

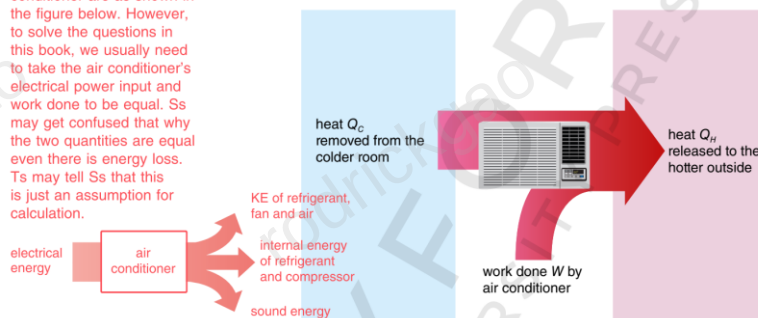


Fig 1.3e  $Q_H = Q_C + W$ .

The heat delivered outside equals heat removed from the cold space plus electrical work input:  $Q_H = Q_C + W$ . The original teacher annotations are retained at the left.

### E3.1.5

$$P_{\text{cool}} = \frac{Q_C}{t}$$

Cooling capacity  $P_{\text{cool}}$  (W);  $Q_C$ : heat removed from the cold reservoir (J);  $t$ : s. Not the electrical input power.

⚡ Ts may tell Ss that the energy conversions of an air conditioner are as shown in the figure below. However, to solve the questions in this book, we usually need to take the air conditioner's electrical power input and work done to be equal. Ss may get confused that why the two quantities are equal even there is energy loss. Ts may tell Ss that this is just an assumption for calculation.

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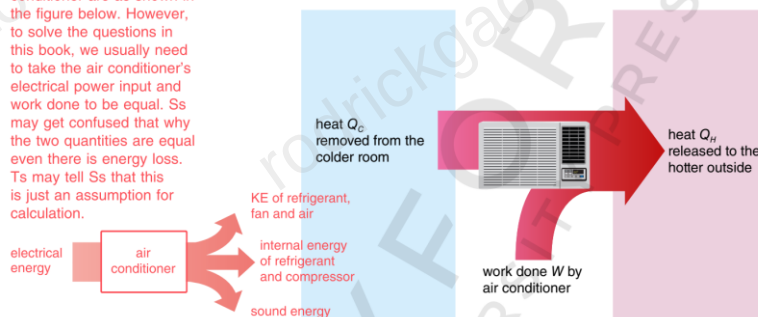


Fig 1.3e  $Q_H = Q_C + W$ .

The heat delivered outside equals heat removed from the cold space plus electrical work input:  $Q_H = Q_C + W$ . The original teacher annotations are retained at the left.

### E3.1.6

$$\text{COP}_{\text{cool}} = \frac{P_{\text{cool}}}{P_{\text{in}}} = \frac{Q_C}{W}$$

Cooling coefficient of performance is dimensionless and may exceed 1.  $Q_C$ : heat removed;  $W$ : work input, both in J and over the same interval.

Ts may tell Ss that the energy conversions of an air conditioner are as shown in the figure below. However, to solve the questions in this book, we usually need to take the air conditioner's electrical power input and work done to be equal. Ss may get confused that why the two quantities are equal even there is energy loss. Ts may tell Ss that this is just an assumption for calculation.

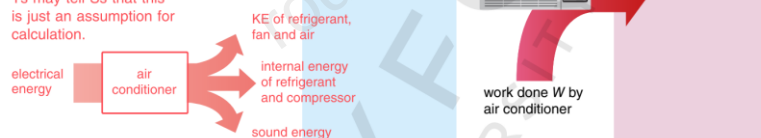


Fig 1.3e  $Q_H = Q_C + W$

In other words, the amount of heat released to the outside is greater than the heat removed from the room, i.e.  $Q_H > Q_C$ .

The heat delivered outside equals heat removed from the cold space plus electrical work input:  $Q_H = Q_C + W$ . The original teacher annotations are retained at the left.

## E302 Lighting

### E3.2.1

Luminous flux is the power output of a light source corrected for the dependence on wavelength of the sensitivity of the human eye.

### E3.2.2

$$E = \frac{\Phi}{A}$$

$E$ : illuminance (lx);  $\Phi$ : luminous flux received by the surface (lm);  $A$ : illuminated area ( $\text{m}^2$ ). Uniform illumination, or average illuminance over  $A$ .

### E3.2.3

Luminous flux measures the power emitted by a source while illuminance measures the power per unit area received by a surface, both corrected for the dependence on wavelength of the sensitivity of the human eye.

### E3.2.4

$$E = E_{\perp} \cos \theta = \frac{\Phi \cos \theta}{4\pi r^2}$$

Isotropic point source:  $\Phi$ : total emitted luminous flux (lm);  $r$ : source-to-surface distance (m);  $\theta$ : angle between incident light and surface normal;  $E$ : lx.

### E3.2.5

$$\eta_v = \frac{\Phi}{P_{in}}$$

Luminous efficacy  $\eta_v$  (lm W<sup>-1</sup>);  $\Phi$ : useful luminous flux (lm);  $P_{in}$ : input electrical power (W). Not a dimensionless energy efficiency.

### E3.2.6

$$E = \frac{\Phi}{A}$$

E: illuminance (lx);  $\Phi$ : luminous flux received by the surface (lm); A: illuminated area (m<sup>2</sup>). Uniform illumination, or average illuminance over A.

### E3.2.7

$$\eta_v = \frac{\Phi}{P_{in}}$$

Luminous efficacy  $\eta_v$  (lm W<sup>-1</sup>);  $\Phi$ : useful luminous flux (lm);  $P_{in}$ : input electrical power (W). Not a dimensionless energy efficiency.

## E303 Energy Efficiency

### E3.3.1

$$\frac{Q}{t} = \frac{\kappa A(T_H - T_C)}{d}$$

Steady conduction through a uniform slab.  $\kappa$ : thermal conductivity (W m<sup>-1</sup> K<sup>-1</sup>); A: m<sup>2</sup>; d: thickness (m); temperature difference: K or °C; Q/t: W.

### E3.3.2

$$U = \frac{\kappa}{d}$$

Single homogeneous layer, neglecting surface resistances. U: thermal transmittance (W m<sup>-2</sup> K<sup>-1</sup>);  $\kappa$ : W m<sup>-1</sup> K<sup>-1</sup>; d: thickness (m).

### E3.3.3

$$OTTV = \frac{Q_C/t + Q_S/t}{A_T}$$

Overall thermal transfer value (W m<sup>-2</sup>);  $Q_C/t$ : conductive heat gain (W);  $Q_S/t$ : solar heat gain (W);  $A_T$ : total area of the building envelope being considered (m<sup>2</sup>).

#### E3.3.4

During braking, a regenerative braking system converts the kinetic energy of a vehicle into electrical energy, which is then stored as chemical energy by the batteries.

### E304 Energy and Energy Consumption

#### E3.4.1

The binding energy of a nucleus is the amount of energy required to break a nucleus into its constituent nucleons.

#### E3.4.2

$$\frac{E_b}{\text{MeV}} \approx 931 \frac{\Delta m}{\text{u}}$$

$E_b$ : binding energy;  $\Delta m$ : mass of separated nucleons minus nuclear mass. 1 u corresponds to approximately 931 MeV/c<sup>2</sup>.

#### E3.4.3

$$P = \frac{1}{2} \eta \rho A v^3$$

Wind turbine output  $P$  (W);  $\eta$ : fractional efficiency; air density  $\rho$  (kg m<sup>-3</sup>); swept rotor area  $A$  (m<sup>2</sup>); wind speed  $v$  (m s<sup>-1</sup>). Uniform wind through  $A$ .

#### E3.4.4

$$P = \eta \frac{mgh}{t}$$

Hydroelectric output  $P$  (W);  $\eta$ : fractional efficiency;  $m/t$ : mass flow rate (kg s<sup>-1</sup>);  $g$ : m s<sup>-2</sup>;  $h$ : vertical head (m).

#### E3.4.5

The solar constant is the total electromagnetic radiation energy radiated at normal incidence by the Sun per unit time per unit area at the mean distance between the Earth and the Sun, measured outside the Earth's atmosphere.

#### E3.4.6

The greenhouse effect is a natural process by which the atmosphere of the Earth traps some of the thermal radiation from the Earth's surface and warms the Earth.

# Elective Book 2

## E201 Particle Nature of Light

### E2.1.1

The photoelectric effect is the emission of electrons from the surface of a material exposed to electromagnetic radiation.

### E2.1.2

The stopping potential is the magnitude of the potential difference that is just enough to stop the photoelectrons with the maximum kinetic energy from reaching the anode.

### E2.1.3

$$K_{\max} = eV_s$$

$K_{\max}$ : maximum photoelectron kinetic energy (J);  $V_s$ : stopping potential magnitude (V);  $e = 1.60 \times 10^{-19}$  C, positive elementary charge.

### E2.1.4

The threshold frequency is the minimum frequency of light for the photoelectric effect to occur.

### E2.1.5

$$E = hf = \frac{hc}{\lambda}$$

Photon energy  $E$  (J);  $h = 6.63 \times 10^{-34}$  J s;  $f$ : Hz;  $c \approx 3.00 \times 10^8$  m s<sup>-1</sup>;  $\lambda$ : vacuum wavelength (m).

### E2.1.6

Work function ( $\phi$ ) is the minimum energy required to release an electron from the surface of a material.

### E2.1.7

$$\phi = hf_0$$

$\phi$ : work function (J);  $f_0$ : threshold frequency (Hz);  $h = 6.63 \times 10^{-34}$  J s. Use consistent energy units.

### E2.1.8

$$K_{max} = hf - \phi = h(f - f_0)$$

Photoemission requires  $f \geq f_0$ .  $K_{max}$  and  $\phi$ : J;  $f$ ,  $f_0$ : Hz;  $h = 6.63 \times 10^{-34}$  J s. A negative result means no photoemission, not negative kinetic energy.

### E2.1.9

$$V_s = \frac{h}{e} f - \frac{\phi}{e}$$

For  $f \geq f_0$ :  $V_s$  is stopping-potential magnitude (V); slope  $h/e$ ; intercept  $-\phi/e$ .  $h = 6.63 \times 10^{-34}$  J s;  $e = 1.60 \times 10^{-19}$  C;  $\phi$ : J.

### E2.1.10

Light exhibits wave-particle duality.

## E202 Towards a Modern Atomic Model

### E2.2.1

Rutherford's nuclear model concentrates all positive charge and nearly all atomic mass in a tiny nucleus, with electrons outside it. The classical orbit picture uses electric attraction as centripetal force, but cannot explain atomic stability; Bohr's model introduces quantized states.

### E2.2.2

$$m_e v_n r_n = \frac{nh}{2\pi}$$

Bohr hydrogen model:  $m_e$ : electron mass (kg);  $v_n$ : orbital speed ( $\text{m s}^{-1}$ );  $r_n$ : radius (m);  $n$ : positive integer;  $h$ : Planck constant (J s).

### E2.2.3

$$L = \frac{nh}{2\pi}, n = 1, 2, 3, \dots$$

Bohr hydrogen model: electric attraction supplies centripetal force for circular orbits. Only discrete stationary orbits are allowed; an electron in one such orbit does not radiate and has constant energy. Transitions between orbits involve absorption or emission of a photon with the energy difference. Orbital angular momentum  $L$  (J s) is quantized;  $h = 6.63 \times 10^{-34}$  J s.

### E2.2.4

$$r_n = n^2 r_1$$

Bohr hydrogen atom:  $r_n$ : radius of orbit  $n$ ;  $r_1$ : ground-state radius;  $n = 1, 2, 3, \dots$

| State             | Quantum number | Energy level                                 | Radius of orbit                                                      |
|-------------------|----------------|----------------------------------------------|----------------------------------------------------------------------|
| ground state      | $n = 1$        | $E_1 = -13.6 \text{ eV}$                     | $R_1 = 5.29 \times 10^{-11} \text{ m}$                               |
| 1st excited state | $n = 2$        | $E_2 = -\frac{13.6}{2^2} = -3.40 \text{ eV}$ | $R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10} \text{ m}$ |
| 2nd excited state | $n = 3$        | $E_3 = -\frac{13.6}{3^2} = -1.51 \text{ eV}$ | $R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10} \text{ m}$ |
| 3rd excited state | $n = 4$        | $E_4 = -\frac{13.6}{4^2} = -0.85 \text{ eV}$ | $R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10} \text{ m}$ |

**Table 2.2b** Energy levels and radii of orbits in different states.

As quantum number  $n$  increases, bound-state energy approaches zero from below and orbit radius increases. The energy spacing becomes smaller.

### E2.2.5

$$E_n = \frac{E_1}{n^2} = \frac{-13.6 \text{ eV}}{n^2}$$

Hydrogen atom; zero energy is a stationary free electron infinitely far from the nucleus.  $n$ : positive integer;  $E_1 = -13.6 \text{ eV}$ .

| State             | Quantum number | Energy level                                 | Radius of orbit                                                      |
|-------------------|----------------|----------------------------------------------|----------------------------------------------------------------------|
| ground state      | $n = 1$        | $E_1 = -13.6 \text{ eV}$                     | $R_1 = 5.29 \times 10^{-11} \text{ m}$                               |
| 1st excited state | $n = 2$        | $E_2 = -\frac{13.6}{2^2} = -3.40 \text{ eV}$ | $R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10} \text{ m}$ |
| 2nd excited state | $n = 3$        | $E_3 = -\frac{13.6}{3^2} = -1.51 \text{ eV}$ | $R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10} \text{ m}$ |
| 3rd excited state | $n = 4$        | $E_4 = -\frac{13.6}{4^2} = -0.85 \text{ eV}$ | $R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10} \text{ m}$ |

**Table 2.2b** Energy levels and radii of orbits in different states.

As quantum number  $n$  increases, bound-state energy approaches zero from below and orbit radius increases. The energy spacing becomes smaller.

### E2.2.6

$$E_{\text{exc}} = E_n - E_1$$

Excitation from ground state to bound level  $n > 1$ . Energy difference is positive; use consistent energy units.

| State             | Quantum number | Energy level                                 | Radius of orbit                                                      |
|-------------------|----------------|----------------------------------------------|----------------------------------------------------------------------|
| ground state      | $n = 1$        | $E_1 = -13.6 \text{ eV}$                     | $R_1 = 5.29 \times 10^{-11} \text{ m}$                               |
| 1st excited state | $n = 2$        | $E_2 = -\frac{13.6}{2^2} = -3.40 \text{ eV}$ | $R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10} \text{ m}$ |
| 2nd excited state | $n = 3$        | $E_3 = -\frac{13.6}{3^2} = -1.51 \text{ eV}$ | $R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10} \text{ m}$ |
| 3rd excited state | $n = 4$        | $E_4 = -\frac{13.6}{4^2} = -0.85 \text{ eV}$ | $R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10} \text{ m}$ |

**Table 2.2b** Energy levels and radii of orbits in different states.

As quantum number  $n$  increases, bound-state energy approaches zero from below and orbit radius increases. The energy spacing becomes smaller.

### E2.2.7

$$E_{\text{ion}} = E_{\infty} - E_1 = 13.6 \text{ eV}$$

Hydrogen atom initially in its ground state;  $E_{\infty} = 0$ . From excited level  $n$ , ionization requires  $13.6/n^2$  eV.

| State             | Quantum number | Energy level                                 | Radius of orbit                                                      |
|-------------------|----------------|----------------------------------------------|----------------------------------------------------------------------|
| ground state      | $n = 1$        | $E_1 = -13.6 \text{ eV}$                     | $R_1 = 5.29 \times 10^{-11} \text{ m}$                               |
| 1st excited state | $n = 2$        | $E_2 = -\frac{13.6}{2^2} = -3.40 \text{ eV}$ | $R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10} \text{ m}$ |
| 2nd excited state | $n = 3$        | $E_3 = -\frac{13.6}{3^2} = -1.51 \text{ eV}$ | $R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10} \text{ m}$ |
| 3rd excited state | $n = 4$        | $E_4 = -\frac{13.6}{4^2} = -0.85 \text{ eV}$ | $R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10} \text{ m}$ |

**Table 2.2b** Energy levels and radii of orbits in different states.

As quantum number  $n$  increases, bound-state energy approaches zero from below and orbit radius increases. The energy spacing becomes smaller.

### E2.2.8

$$hf = \frac{hc}{\lambda} = E_a - E_b$$

Emission from higher level  $a$  to lower level  $b$ ,  $a > b$ . Photon energy is positive. Convert eV to J when using  $h$  in J s;  $\lambda$  is vacuum wavelength.

### E2.2.9

$$\frac{1}{\lambda} = \frac{13.6 \text{ eV}}{hc} \left( \frac{1}{b^2} - \frac{1}{a^2} \right)$$

Hydrogen emission  $a > b$ . Convert 13.6 eV to joules if  $h$  is in J s and  $c$  in  $\text{m s}^{-1}$ ; then  $\lambda$  is in metres.

### E2.2.10

$$p = \frac{h}{\lambda}$$

Photon momentum  $p$  ( $\text{kg m s}^{-1}$ );  $h = 6.63 \times 10^{-34} \text{ J s}$ ;  $\lambda$ : vacuum wavelength (m).

### E2.2.11

$$\lambda = \frac{h}{p} \approx \frac{h}{mv}$$

de Broglie wavelength  $\lambda$  (m); momentum  $p$  ( $\text{kg m s}^{-1}$ );  $h = 6.63 \times 10^{-34} \text{ J s}$ .  $p \approx mv$  only at non-relativistic speed.

## E203 Probing into the Nano World

### E2.3.1

$$1\text{nm} = 10^{-9}\text{m}$$

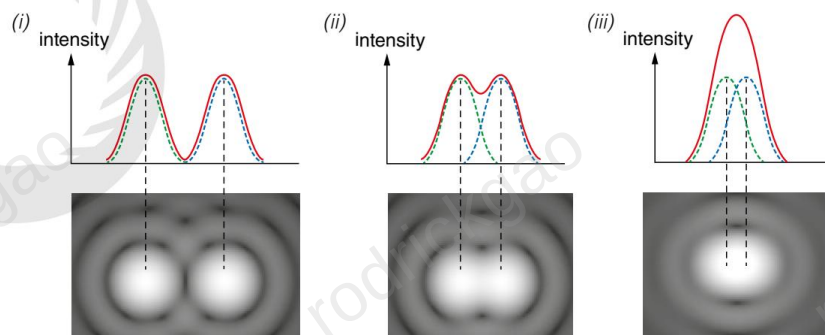
Nanomaterials typically have at least one dimension between 1 and 100 nm.

### E2.3.2

When materials are reduced to nano scale in size, they may exhibit different electrical (e.g. conductivity), mechanical (e.g. strength and hardness) and optical (e.g. colour and transparency) properties.

### E2.3.3

Two points can just be resolved when the central maximum of the diffraction pattern of one point is centred on the first-order minimum of the diffraction pattern of the other.



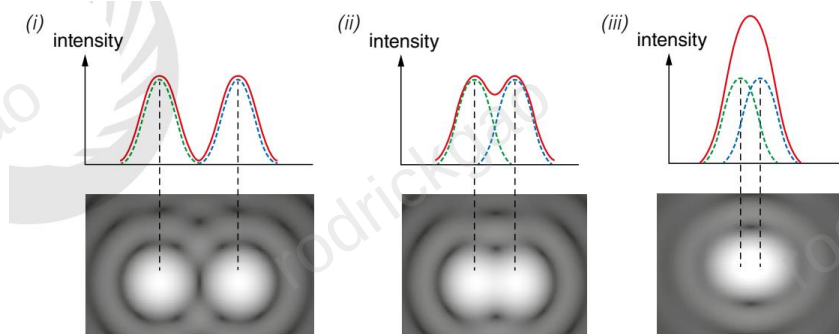
**Fig 3.2d** According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.

Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

### E2.3.4

$$\theta_R \approx 1.22 \frac{\lambda}{d}$$

Rayleigh criterion for a circular aperture.  $\theta_R$ : minimum resolvable angular separation (rad);  $\lambda$ : wavelength (m);  $d$ : aperture diameter (m), or pupil diameter for the eye.



**Fig 3.2d** According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.

Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

### E2.3.5

In a TEM, fast-moving electrons are directed through a thin sample to form an image.

The magnetic lenses produce magnetic fields to converge the electron beam, like a convex lens converging light.

Increasing the accelerating voltage of the electron gun will increase the speeds of the electrons and reduce their de Broglie wavelengths, so the TEM will attain a higher resolving power.

### E2.3.6

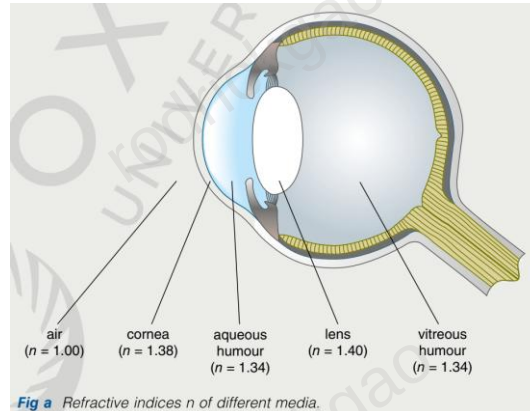
A scanning tunnelling microscope makes use of the tunnelling current to produce an image of a sample surface with details down to atomic scale. It uses a probe to scan across the sample surface either at a constant height or with a constant current.

# Elective Book 4

## E401 Human Vision and Optical Fibre Endoscopy

### E4.1.1

Accommodation is the process of changing the shape (or focal length) of the lens to form a sharp image on the retina.



The cornea and lens refract light; accommodation changes the lens shape while the retina distance remains almost fixed.

### E4.1.2

Rods are responsible for black and white vision and vision in low light conditions.

Cones are responsible for colour vision. They are only active at levels of high brightness.

### E4.1.3

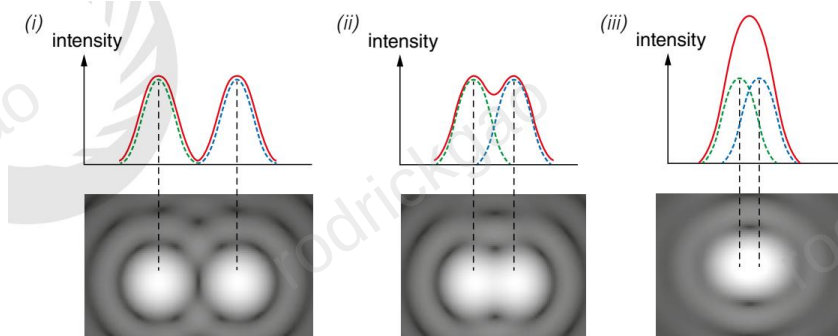
$$\theta \approx \frac{x}{u}$$

Small visual angle  $\theta$  in radians;  $x$ : object size;  $u$ : viewing distance, both in the same length unit.

#### E4.1.4

$$\theta_R \approx 1.22 \frac{\lambda}{d}$$

Rayleigh criterion for a circular aperture.  $\theta_R$ : minimum resolvable angular separation (rad);  $\lambda$ : wavelength (m);  $d$ : aperture diameter (m), or pupil diameter for the eye.



**Fig 3.2d** According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.

Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

#### E4.1.5

$$P = \frac{1}{f}$$

Optical power  $P$  in dioptres ( $D = m^{-1}$ ); focal length  $f$  MUST be in metres. Converging lens: positive; diverging lens: negative.

#### E4.1.6

$$P_{\text{eff}} \approx P_1 + P_2 + \dots$$

Thin lenses in contact or with negligible separation; powers in dioptres. Significant separation requires an additional correction.

#### E4.1.7

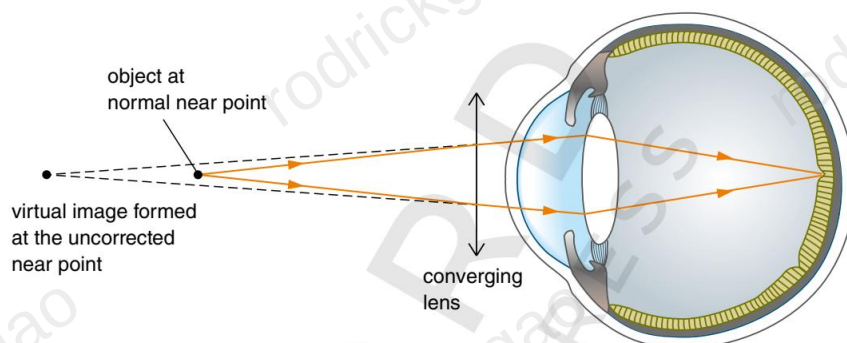
Near point: the closest point at which an object can still be seen sharply; conventionally about 25 cm for a normal eye.

Far point: the furthest point at which an object can be seen sharply with a relaxed eye; at infinity for a normal eye.

### E4.1.8

Common vision defects and their correction:

- Short-sightedness: distant objects are blurred because the eyeball is too long or the eye lens is too powerful; correct with a diverging lens.
- Long-sightedness: near objects are blurred because the eyeball is too short or the eye lens is too weak; correct with a converging lens.
- Presbyopia (old-sightedness): near objects are blurred because the eye lens loses elasticity with age; correct with a converging lens.



**Fig 1.2h** Correcting old-sightedness with a converging lens.

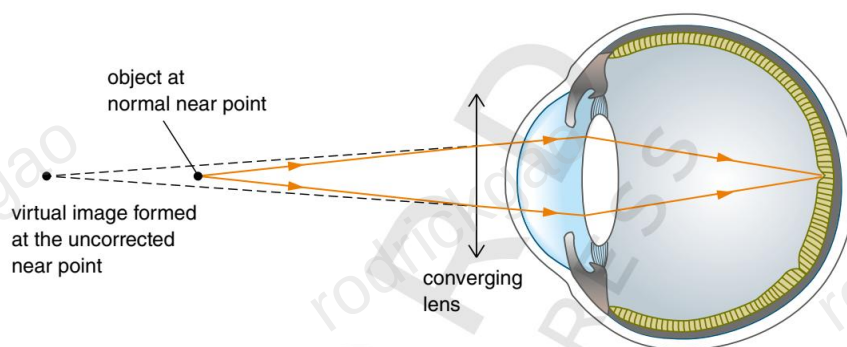
The converging spectacle lens produces a virtual image at the unaided near point. The eye then forms a real image on the retina; these are two different images.

### E4.1.9

$$P_{\text{spectacles}} = \frac{1}{u} + \frac{1}{v}$$

$$P_{\text{spectacles}} = \frac{1}{d_{\text{corrected}}} - \frac{1}{d_{\text{uncorrected}}}$$

All distances in metres; power in D. Real object  $u = d_{\text{corrected}} > 0$ ; virtual image at the unaided near/far point has  $v = -d_{\text{uncorrected}}$ . Use  $1/\infty = 0$ . Lens-eye separation neglected.



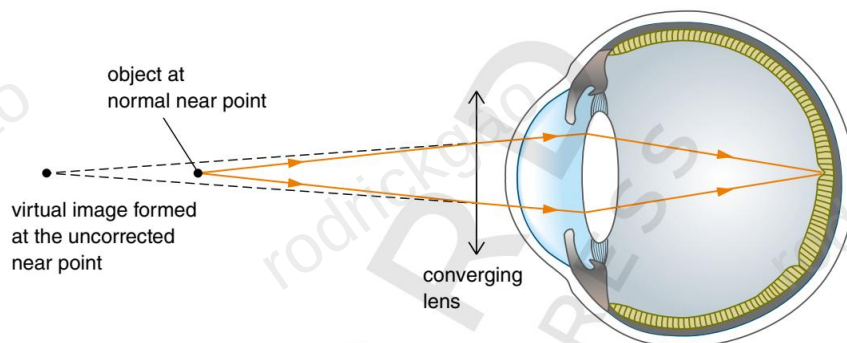
**Fig 1.2h** Correcting old-sightedness with a converging lens.

The converging spectacle lens produces a virtual image at the unaided near point. The eye then forms a real image on the retina; these are two different images.

#### E4.1.10

$$P_{\text{eye}} = \frac{1}{u} + \frac{1}{v}$$

Unaided eye model:  $u$ : real-object distance at near/far point (m), positive;  $v$ : optical-centre-to-retina distance (m), positive;  $P$ : D.



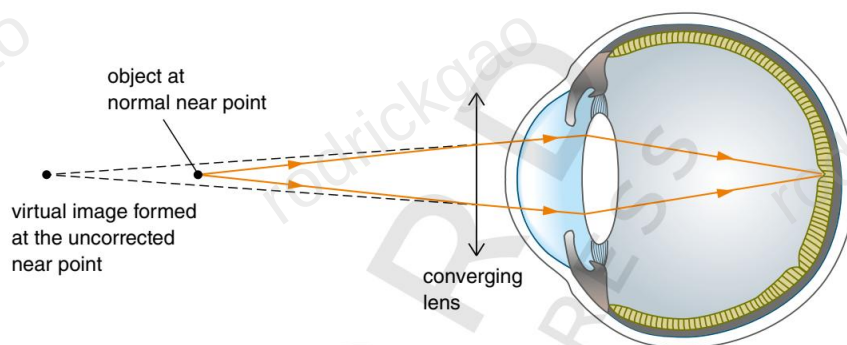
**Fig 1.2h** Correcting old-sightedness with a converging lens.

The converging spectacle lens produces a virtual image at the unaided near point. The eye then forms a real image on the retina; these are two different images.

#### E4.1.11

$$P_{\text{eff}} = \frac{1}{u} + \frac{1}{v}$$

Corrected eye plus spectacles treated as a combined thin lens:  $u$ : corrected near/far-point distance (m);  $v$ : retina distance (m);  $P_{\text{eff}}$ : D. Real object/image distances positive.



**Fig 1.2h** Correcting old-sightedness with a converging lens.

The converging spectacle lens produces a virtual image at the unaided near point. The eye then forms a real image on the retina; these are two different images.

#### E4.1.12

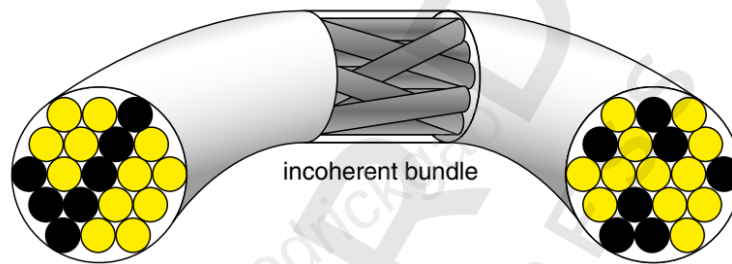
$$n_1 \sin C = n_2 \sin 90^\circ$$

$$\sin C = \frac{n_2}{n_1}$$

Core refractive index  $n_1 >$  cladding index  $n_2$ ;  $C$ : critical angle from the normal. Total internal reflection requires incidence angle greater than  $C$ .

### E4.1.13

- A coherent bundle fibre preserves the relative position of each fibre from one end to the other. It is used to transmit images.
- An incoherent bundle fibre is used to transmit light for illumination.



**Fig 1.3j** An incoherent bundle fibre cannot be used to transmit images.

An incoherent fibre bundle scrambles relative positions and carries illumination. Image transmission requires a coherent bundle with positions preserved.

## E402 Human Hearing and Ultrasound Scanning

### E4.2.1

Pressure amplification in the middle ear is a combined effect of

- the lever action of the three ear bones (or ossicles) and
- the area ratio of the eardrum to the oval window (with the area of the eardrum much larger than that of the oval window).



**Fig 2.1d** Combined effect of the lever action and the area ratio of the eardrum to the oval window.

Lever action increases force and the smaller oval-window area further increases pressure. The illustrated factors multiply to give the pressure gain.

### E4.2.2

The cochlea acts as a frequency analyser. Regions closer to its base vibrate with peak amplitudes for high frequency incoming sounds while regions closer to its apex vibrate with peak amplitudes for low frequency incoming sounds.

#### E4.2.3

$$I = \frac{E}{At} = \frac{P}{A}$$

I: average sound intensity ( $\text{W m}^{-2}$ ); E: sound energy (J); A: perpendicular area ( $\text{m}^2$ ); t: time (s); P: sound power (W).

#### E4.2.4

$$I = \frac{P}{4\pi r^2}$$

Isotropic point source in a free field, negligible absorption. I:  $\text{W m}^{-2}$ ; acoustic power P: W; distance r: m.

#### E4.2.5

$$L = 10\log_{10}\left(\frac{I}{I_0}\right)$$

$$I_0 = 10^{-12}\text{Wm}^{-2}$$

Sound intensity level L (dB); I and  $I_0$  in identical units. Base-10 logarithm; the ratio inside the logarithm is dimensionless.

#### E4.2.6

A hearing loss is represented by an upward shift in the threshold of hearing.

#### E4.2.7

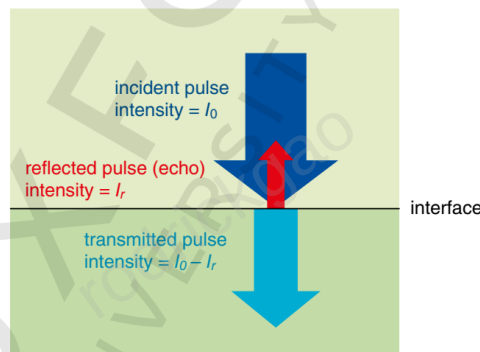
When a fluctuating voltage is applied across the piezoelectric crystal in an ultrasound transducer, the crystal contracts and expands to produce an ultrasound pulse.

When an ultrasound pulse hits the crystal, the crystal is compressed and extended to produce a series of voltage across it.

### E4.2.8

$$Z = \rho c$$

Acoustic impedance  $Z$  ( $\text{kg m}^{-2} \text{s}^{-1}$ ); density  $\rho$  ( $\text{kg m}^{-3}$ ); sound speed  $c$  in the medium ( $\text{m s}^{-1}$ ). Here  $c$  is not light speed.



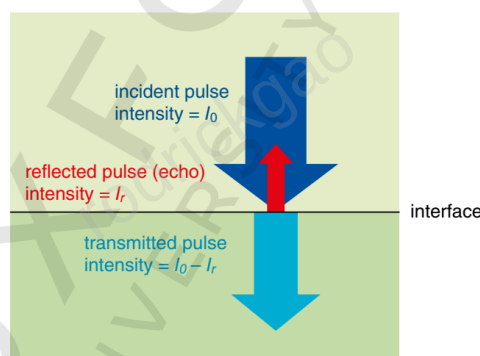
**Fig 2.2h** Reflection and transmission of ultrasound pulse.

Incident intensity divides into reflected and transmitted parts. A larger impedance mismatch gives a larger reflected fraction at normal incidence.

### E4.2.9

$$\alpha = \frac{I_r}{I_i}$$

$\alpha$ : intensity reflection coefficient (dimensionless);  $I_r$ : reflected intensity;  $I_i$ : incident intensity, called  $I_0$  in the textbook. Both intensities in  $\text{W m}^{-2}$ .



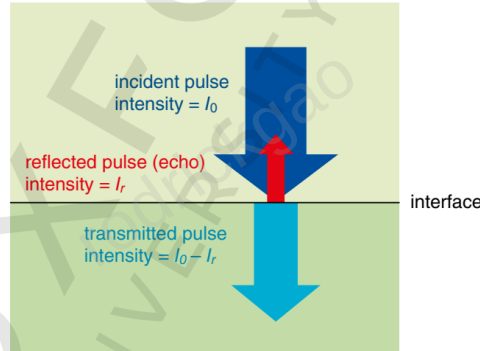
**Fig 2.2h** Reflection and transmission of ultrasound pulse.

Incident intensity divides into reflected and transmitted parts. A larger impedance mismatch gives a larger reflected fraction at normal incidence.

#### E4.2.10

$$\alpha = \frac{I_r}{I_i} = \left( \frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$$

Normal incidence at a plane boundary between two media;  $Z_1$ ,  $Z_2$ : acoustic impedances in the same units.  $\alpha$  is the reflected INTENSITY fraction, not amplitude.



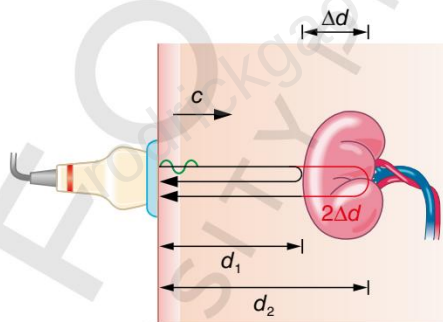
**Fig 2.2h** Reflection and transmission of ultrasound pulse.

Incident intensity divides into reflected and transmitted parts. A larger impedance mismatch gives a larger reflected fraction at normal incidence.

#### E4.2.11

$$\Delta d = \frac{1}{2} c \Delta t$$

Pulse–echo:  $\Delta t$  is the difference in round-trip echo times (s);  $c$ : sound speed ( $\text{m s}^{-1}$ );  $\Delta d$ : separation along the beam (m). Factor  $\frac{1}{2}$  accounts for outward and return paths.



**Fig 2.3d** An ultrasound pulse is reflected from the front and back surfaces of a kidney.

The back-surface echo travels an extra outward and return distance of  $2\Delta d$ . Therefore the time difference gives  $\Delta d = c\Delta t/2$ .

#### E4.2.12

$$f \uparrow \Rightarrow \text{attenuation} \uparrow \Rightarrow \text{penetration} \downarrow$$

In the same tissue, higher ultrasound frequency generally increases attenuation and reduces penetration depth.

### E4.2.13

$$\Delta d_{\min} \approx \frac{\text{spatial pulse length}}{2}$$

Shorter ultrasound pulses improve axial resolution. At a fixed number of cycles per pulse, higher frequency gives shorter pulses and a smaller minimum resolvable separation.

## E403 Medical Imaging Using Ionizing Radiation

### E4.3.1

$$I = I_0 e^{-\mu x}$$

Uniform medium and monoenergetic narrow beam:  $I_0$ : incident intensity;  $I$ : transmitted intensity;  $\mu$ : linear attenuation coefficient;  $x$ : thickness.  $\mu x$  must be dimensionless (e.g.  $\text{cm}^{-1}$  with  $\text{cm}$ ).

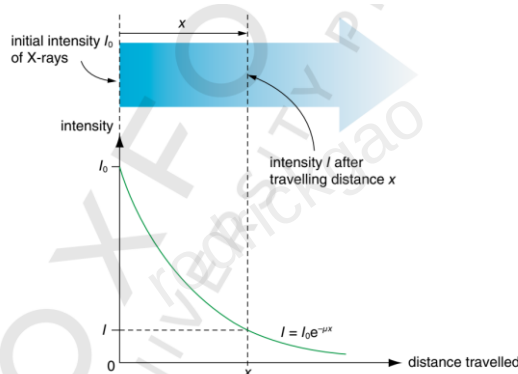


Fig 3.1e Change in intensity over distance travelled.

Intensity decreases exponentially with thickness. Equal additional thicknesses reduce the intensity by equal fractions, not equal absolute amounts.

### E4.3.2

$$x_{1/2} = \frac{\ln 2}{\mu}$$

$x_{1/2}$ : half-value thickness;  $\mu$ : linear attenuation coefficient. Match units, e.g.  $\text{cm}$  and  $\text{cm}^{-1}$ ; natural logarithm.

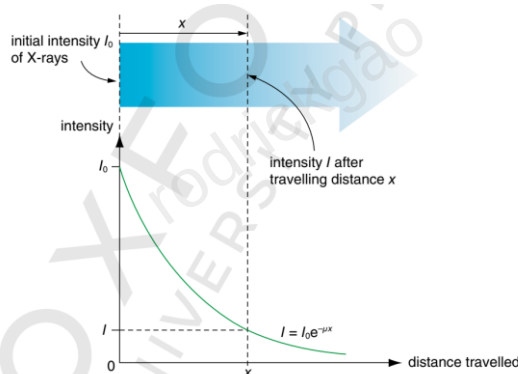


Fig 3.1e Change in intensity over distance travelled.

Intensity decreases exponentially with thickness. Equal additional thicknesses reduce the intensity by equal fractions, not equal absolute amounts.

### E4.3.3

$$\rho \uparrow \Rightarrow \mu \uparrow$$

For otherwise comparable material and photon energy, greater density generally increases linear attenuation. Composition and photon energy also matter.

### E4.3.4

$$E_\gamma \uparrow \Rightarrow \mu \downarrow \Rightarrow \text{penetration} \uparrow$$

General trend over diagnostic X-ray ranges away from absorption edges, for the same material. Not a universal monotonic law across every photon energy.

### E4.3.5

$$I = I_0 e^{-\mu_1 x_1} e^{-\mu_2 x_2} \dots$$

$$I = I_0 e^{-\sum \mu_i x_i}$$

Sequential layers:  $\mu_i$ : layer attenuation coefficient;  $x_i$ : thickness along beam. Each  $\mu_i x_i$  is dimensionless; exponential narrow-beam model.

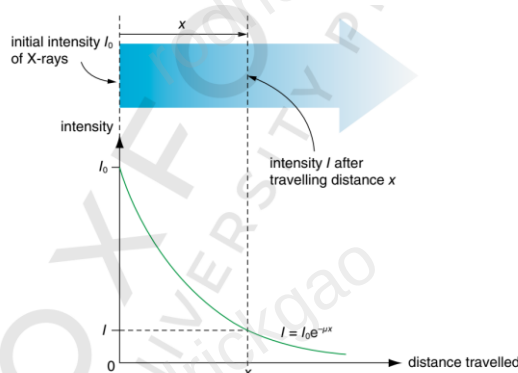


Fig 3.1e Change in intensity over distance travelled.

Intensity decreases exponentially with thickness. Equal additional thicknesses reduce the intensity by equal fractions, not equal absolute amounts.

### E4.3.6

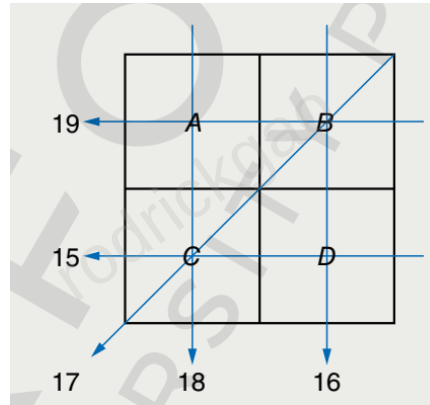
In the conventional textbook X-ray display, regions of high attenuation appear light and regions of low attenuation appear dark. Contrast comes from differences in transmitted intensity; digital display inversion can reverse the light/dark convention.

### E4.3.7

An artificial contrast medium has a high linear attenuation coefficient and can increase the contrast between a body part and its surroundings.

#### E4.3.8

To produce a CT image, the X-ray tube and detector rotate round a slice of a patient's body to obtain multiple attenuation profiles. The profiles are used to produce a map of attenuation coefficients of different body tissues by back projection algorithm. This map of attenuation coefficients is used to generate the CT image.



Each beam records attenuation summed along its path. Measurements at several angles constrain the individual regions A–D used to reconstruct the slice.

#### E4.3.9

In radionuclide imaging, radionuclide tracers are injected into the patient's body and accumulate in the target organ.

The gamma camera detects the  $\gamma$  radiation emitted from the tracers. The data is then processed by a computer to produce a radionuclide image, which is a map of the distribution of the radionuclides in the patient's body.

#### E4.3.10

The physical half-life of a radionuclide is the time taken for half of the nuclei in a sample to decay.

#### E4.3.11

The biological half-life of a substance is the time taken for half of the substance to be removed from the body by biological processes.

#### E4.3.12

$$\frac{1}{t_e} = \frac{1}{t_p} + \frac{1}{t_b}$$
$$t_e = \frac{t_p t_b}{t_p + t_b}$$

Effective half-life  $t_e$ ; physical half-life  $t_p$ ; biological half-life  $t_b$ . Independent exponential processes; all three times in identical units.  $t_e$  is less than either finite component half-life.

#### **E4.3.13**

A cold spot has reduced tracer uptake; a hot spot has increased tracer uptake. In the textbook grayscale convention, cold spots appear lighter and hot spots darker. Other grayscale or colour maps may display them differently.